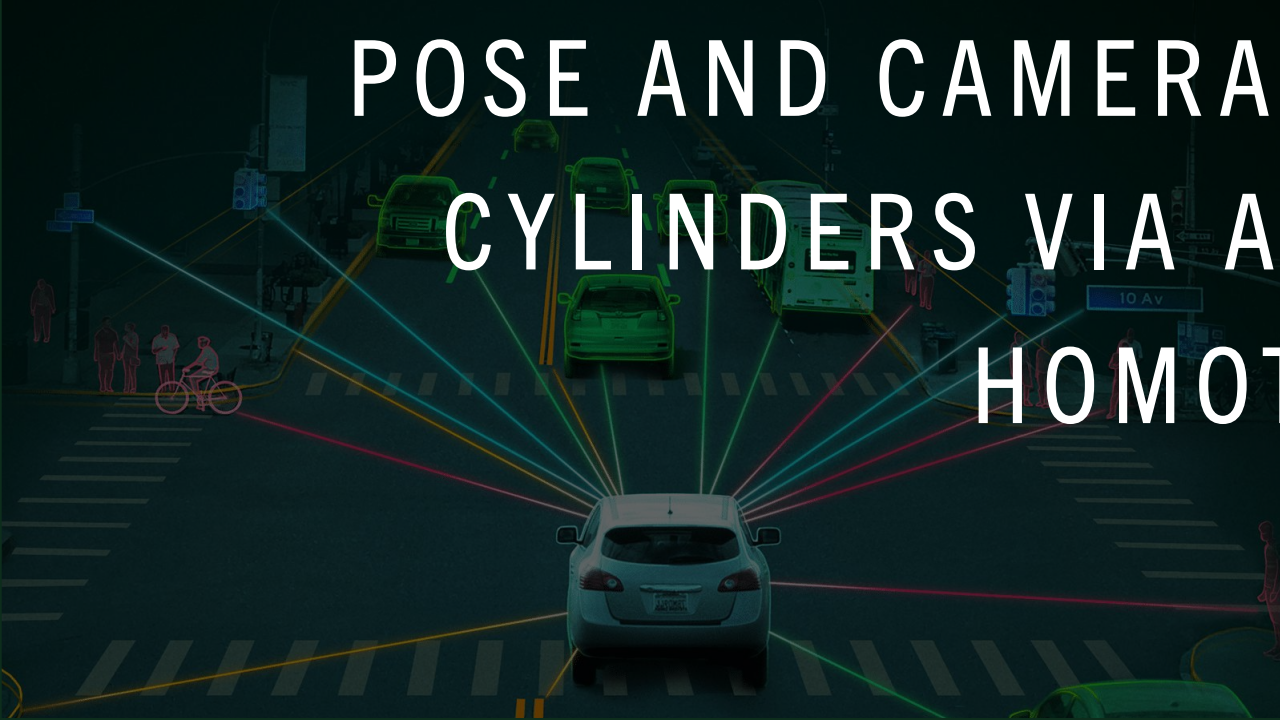




Enrico Barbieri 970131

POSE AND CAMERA CALIBRATION FROM CYLINDERS VIA A SILHOUETTE BASED HOMOTOPY CONTINUATION



Politecnico di Milano

Relatore: Prof. Luca Magri

Correlatore: Prof. Giacomo Boracchi

PROBLEM FORMULATION

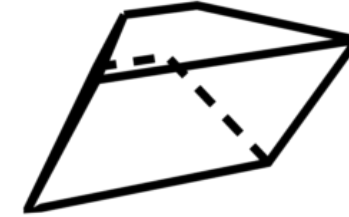
INTUITION

From a known 3D scene composed of cylindrical objects $C_1, \dots, C_M$ we want to estimate each camera pose $[R_1, \mathbf{t}_1], \dots, [R_N, \mathbf{t}_N]$ and the common intrinsic matrix

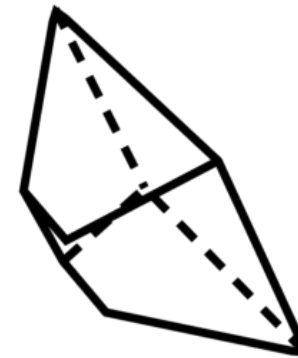
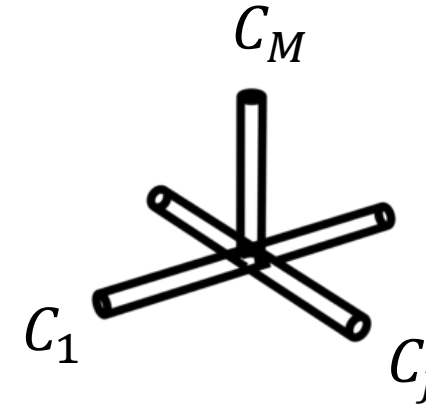
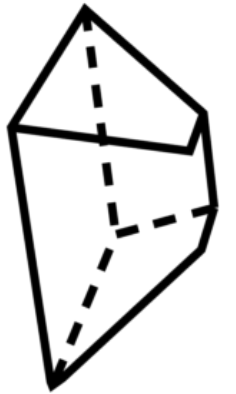
$$K = \begin{bmatrix} f_x & \gamma & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Giving us $P_1, \dots, P_N$ where a generic $P_i = K[R_i | \mathbf{t}_i]$

Camera N ?



Camera i ?



Camera 1 ?

$$K = \begin{bmatrix} f_x & \gamma & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

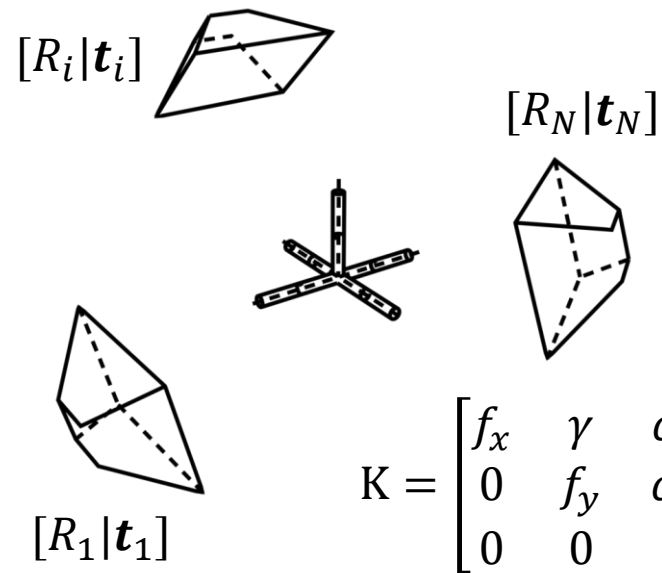
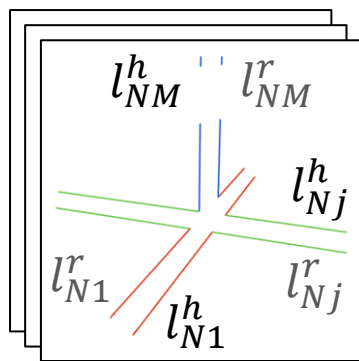
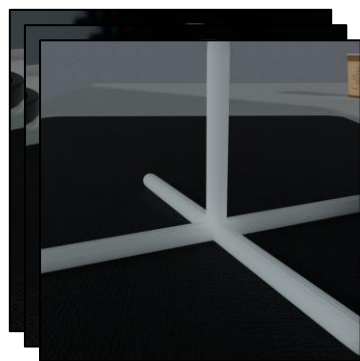
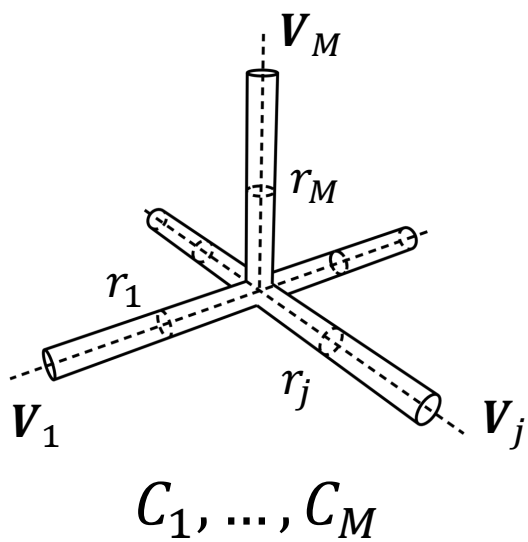
PROBLEM FORMULATION

M cylinders for which direction V_j , radius r_j , and relative position p_j are known.
 N images where the cylinders contours $\mathcal{L} = \{l_{ij}^{(h/r)}\}$ are at least partially visible.

INPUT

Jointly estimating the intrinsic camera matrix K shared by all cameras, and the pose made of rotation R_i and translation t_i for each camera. K is in any possible configuration of unknown parameters $f_x, f_y, \gamma, c_x, c_y$.

OUTPUT



INPUT

INPUT

$\Rightarrow$

OUTPUT

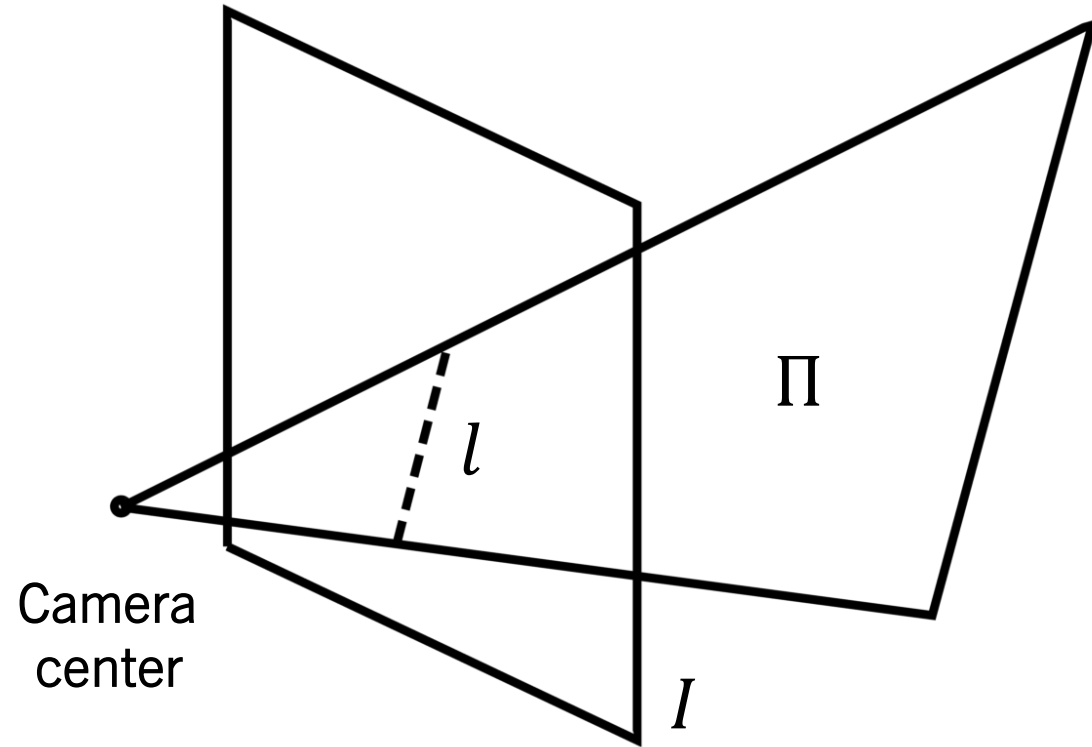
GEOMETRIC CONSTRAINTS

A line l on the image plane forms with the camera center a plane Π such that

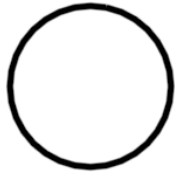
$$\Pi = P^T l \quad (0)$$

A plane Π defines a cylinder surface Q^* if

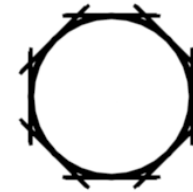
$$\Pi^T Q^* \Pi = 0 \quad (1)$$



GEOMETRIC CONSTRAINTS – DUAL QUADRIC

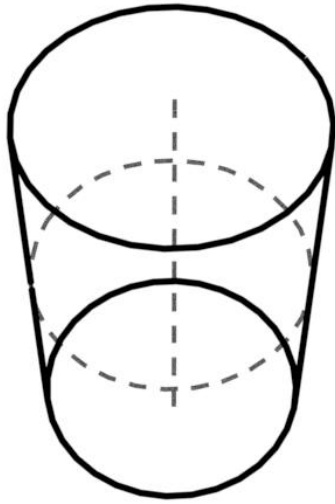


$$\mathbf{x}^T \mathbf{Q} \mathbf{x} = 0$$

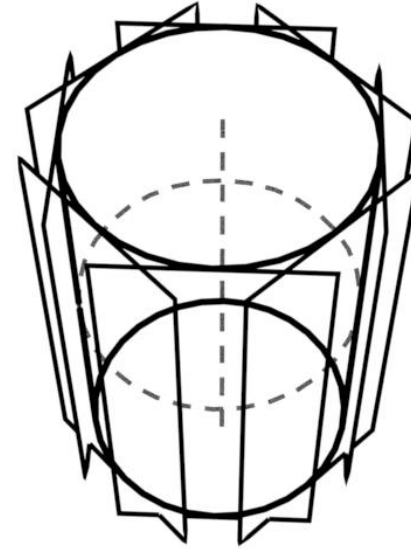


$$\mathbf{l}^T \mathbf{Q}^* \mathbf{l} = 0$$

GEOMETRIC CONSTRAINTS – DUAL QUADRIC

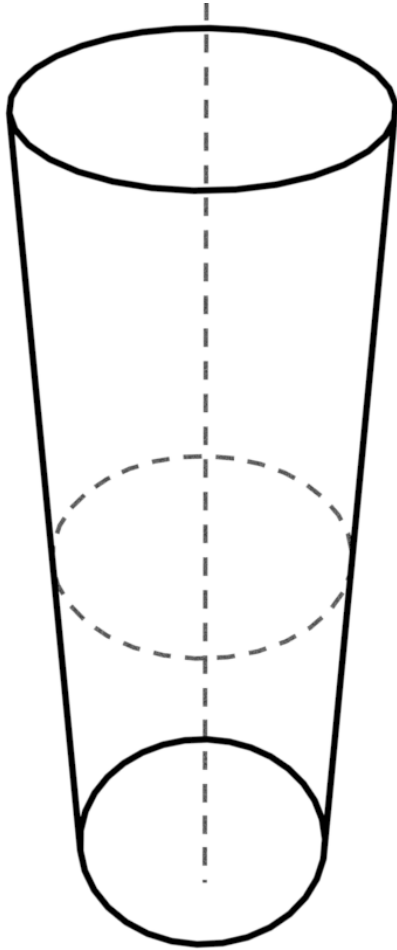


$$X^T Q X = 0$$

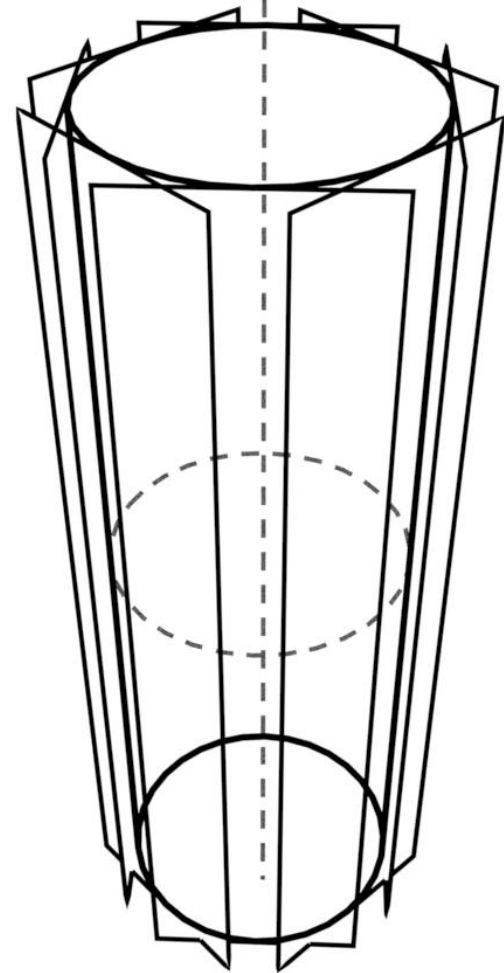


$$H^T Q^* H = 0$$

GEOMETRIC CONSTRAINTS – DUAL QUADRIC



$$X^T Q X = 0$$



$$l^T Q^* l = 0$$

GEOMETRIC CONSTRAINTS

A line l on the image plane forms with the camera center a plane Π such that

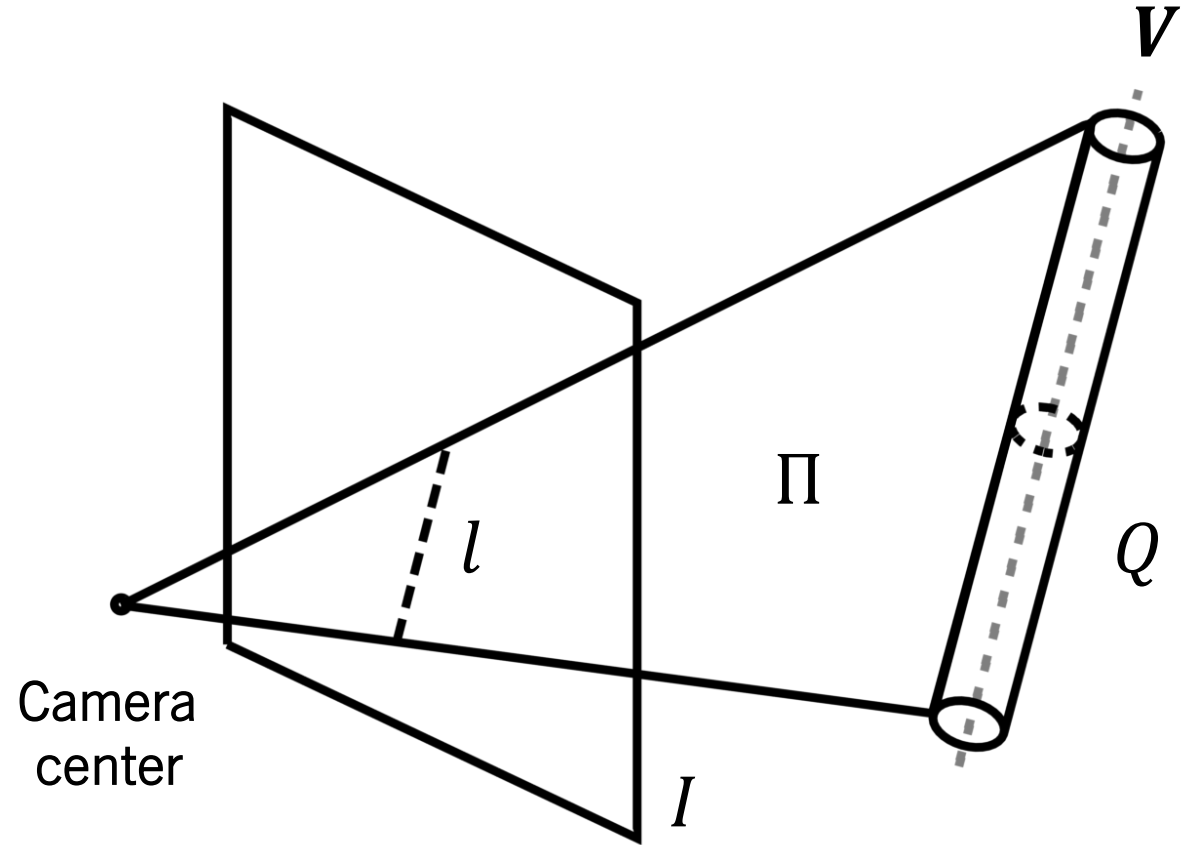
$$\Pi = P^T l \quad (0)$$

A plane Π defines a cylinder surface Q^* if

$$\Pi^T Q^* \Pi = 0 \quad (1)$$

A cylinder surface plane Π passes through the cylinder axis V

$$\Pi^T V = 0 \quad (2)$$



GEOMETRIC CONSTRAINTS

A line l on the image plane forms with the camera center a plane Π such that

$$\Pi = P^T l \quad (0)$$

A plane Π defines a cylinder surface Q^* if

$$\Pi^T Q^* \Pi = 0 \quad (1)$$

A cylinder surface plane Π passes through the cylinder axis V

$$\Pi^T V = 0 \quad (2)$$

$$\begin{cases} l^h{}^T P Q^* P^T l^h = 0 & (1) \\ l^h{}^T P V = 0 & (2) \end{cases}$$

$$\begin{cases} l^h{}^T P Q^* P^T l^h = 0 & (1) \\ l^h{}^T K[R|t] \begin{bmatrix} w \\ 0 \end{bmatrix} = 0 & (2) \end{cases}$$

$$\begin{cases} l^h{}^T K[R|t] Q^* K[R|t]^T l^h = 0 & (1) \\ l^h{}^T K R w = 0 & (2) \end{cases}$$

Two quartic equations for each line: $N \times M \times 2$

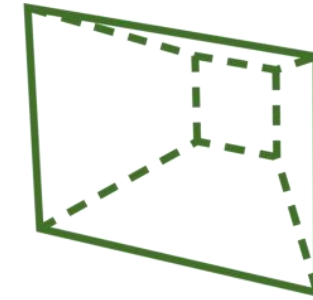
PROPOSED METHOD



HOMOTOPY CONTINUATION

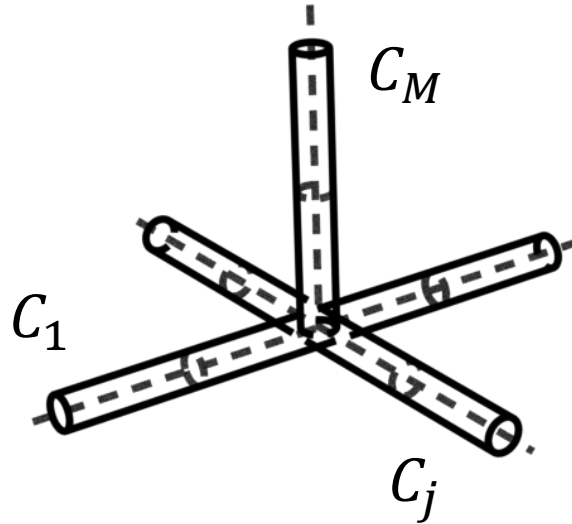
MAIN IDEA

The equations arising from (2) in which $\mathbf{x}_1 = [R_1 | \mathbf{t}_1]$ are quartic. Solving through Gröbner basis requires to lift the problem to a very highly dimensional space.



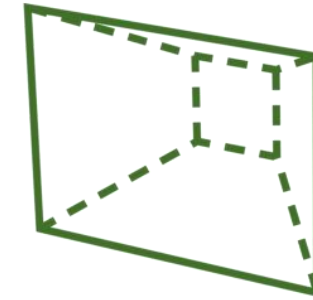
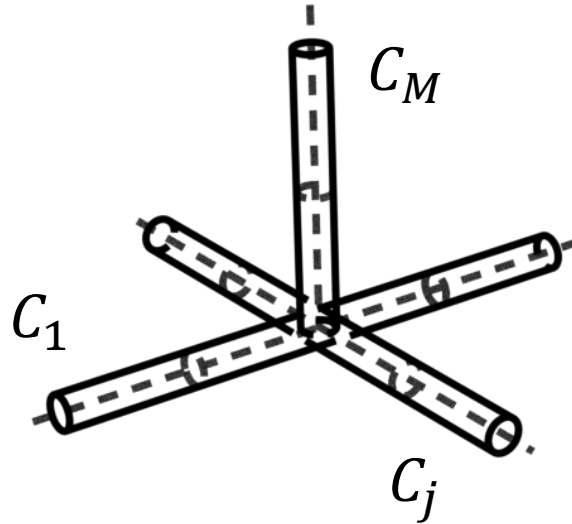
$[R(1) | \mathbf{t}(1)]$

Sought solution



MAIN IDEA

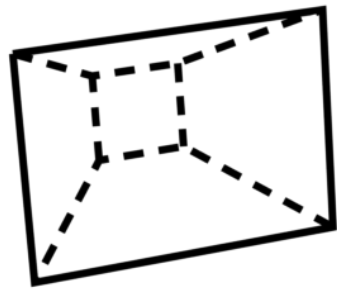
We generate a **synthetic solution** for which we know all parameters, and unknowns and **track how the solution change** when the problem parameters change.



$$[R(1)|t(1)]$$

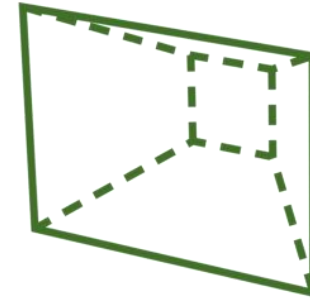
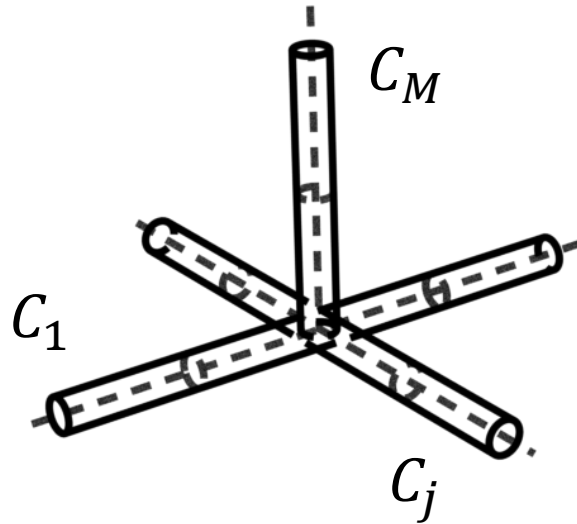
Sought solution

MAIN IDEA



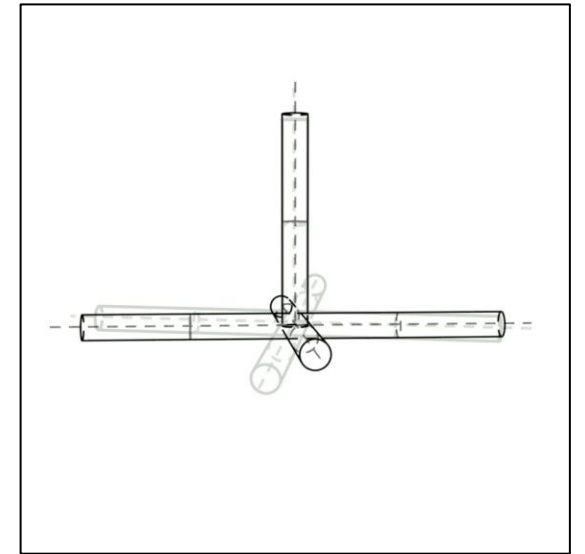
Random solution

$$[R(0)|\mathbf{t}(0)]$$



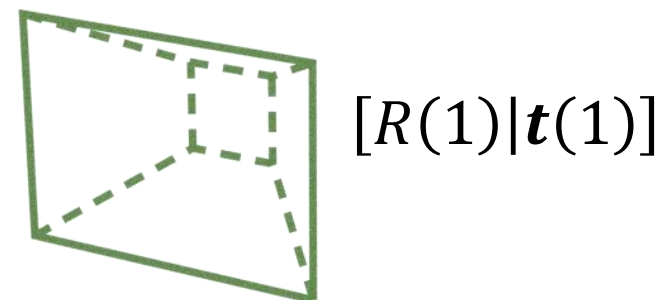
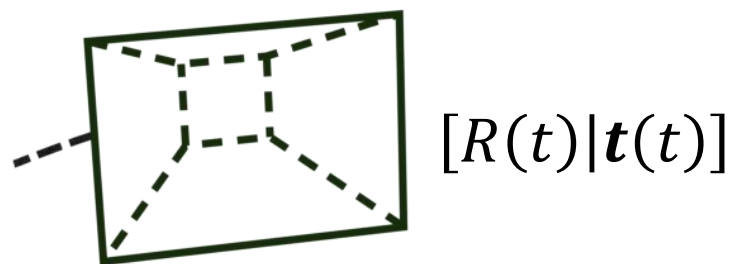
$$[R(1)|\mathbf{t}(1)]$$

Sought solution

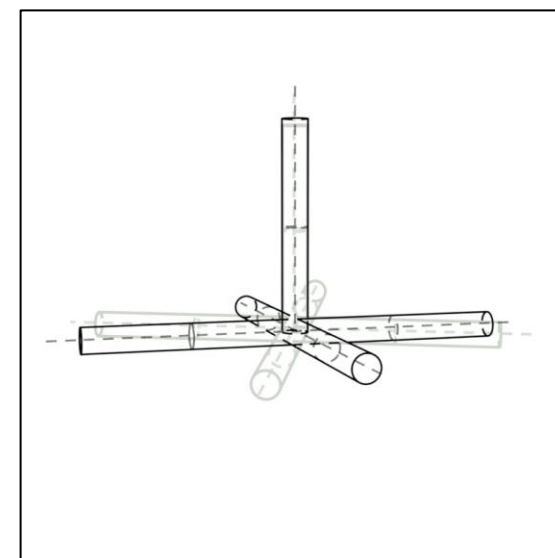
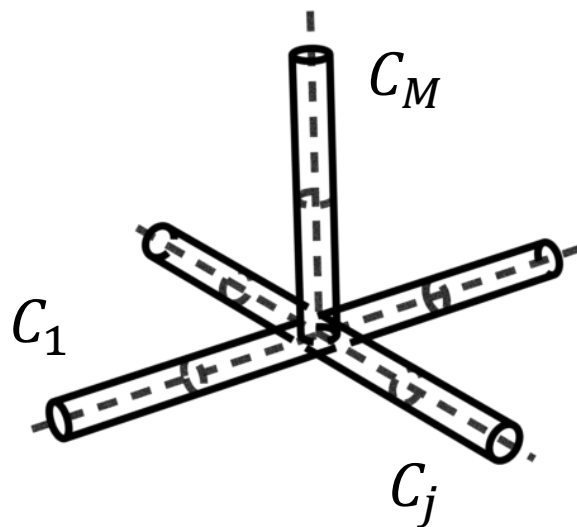


Synth image over real image

MAIN IDEA

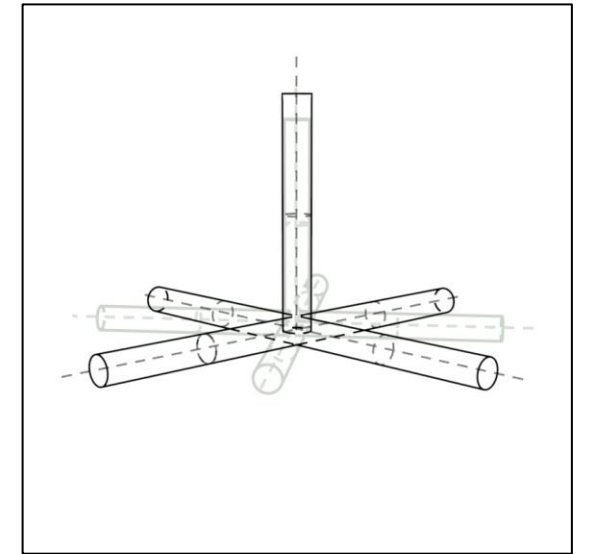
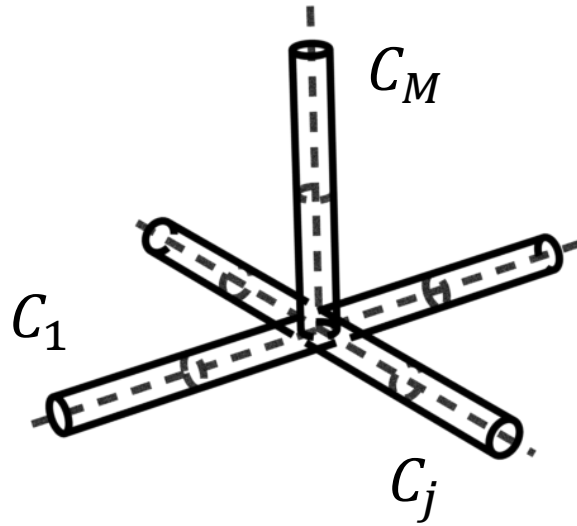
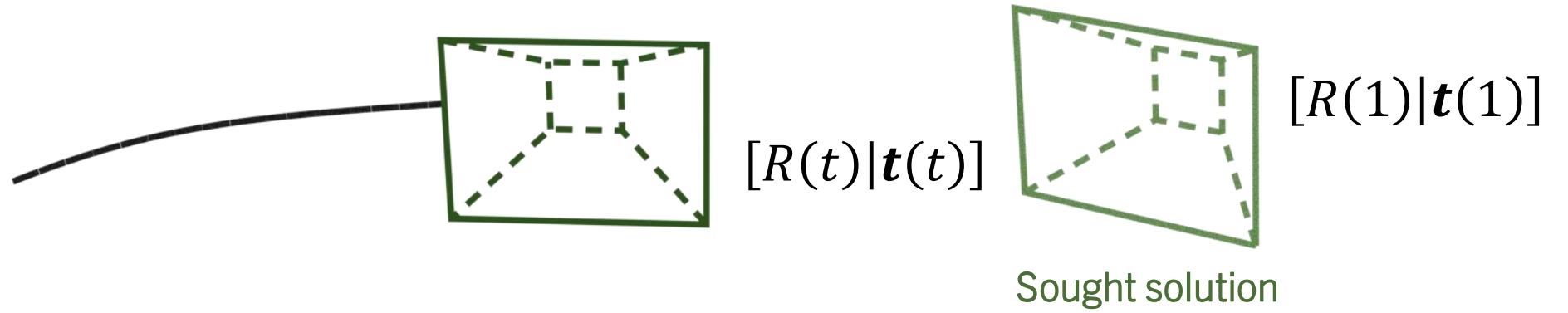


Sought solution



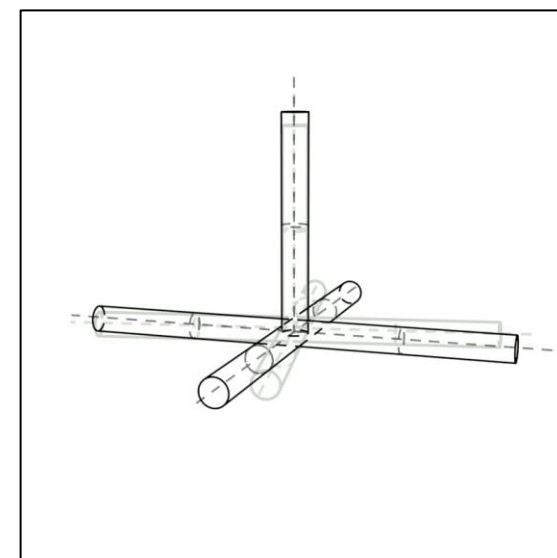
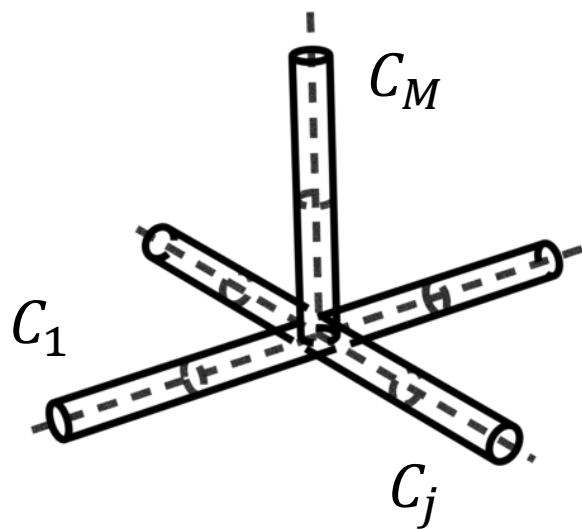
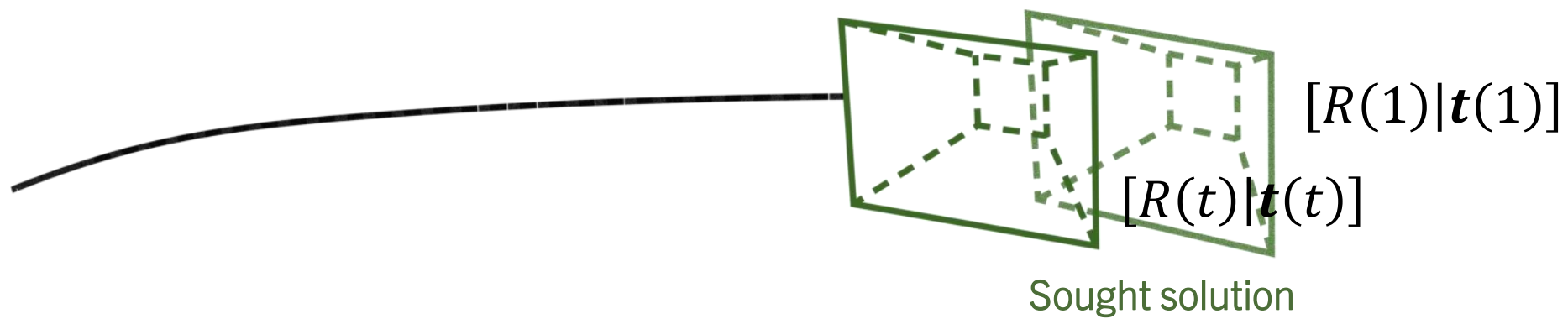
Synth image over real image

MAIN IDEA



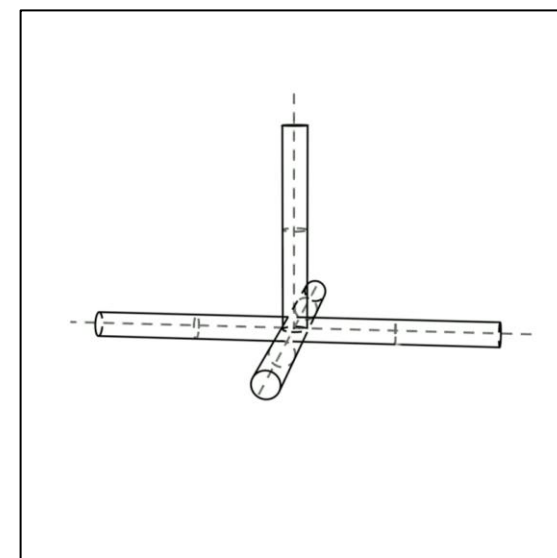
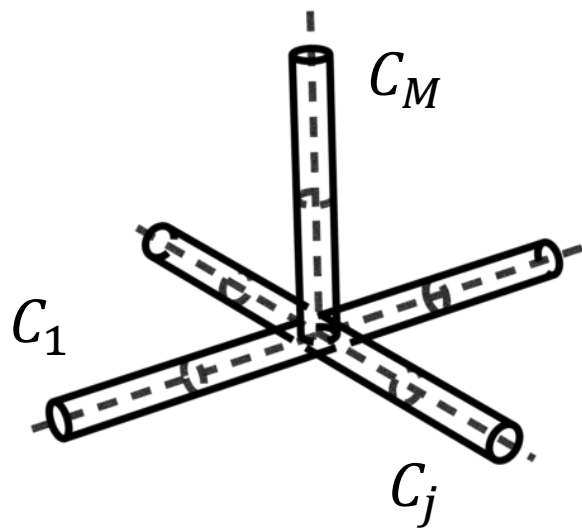
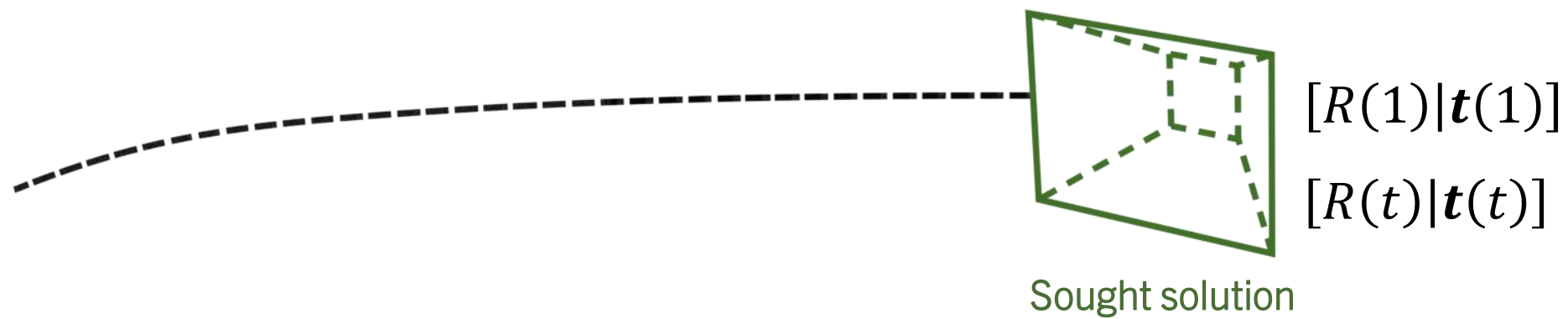
Synth image over real image

MAIN IDEA



Synth image over real image

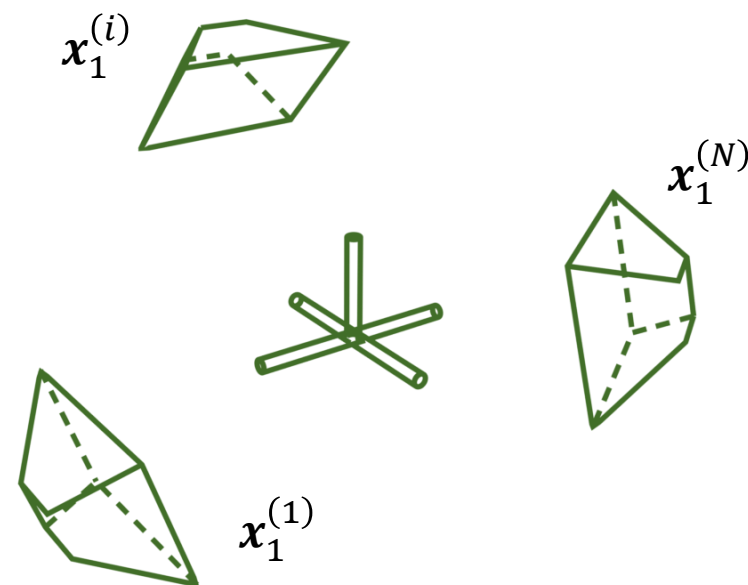
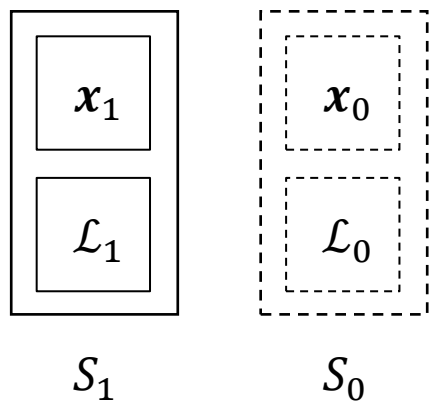
MAIN IDEA



Synth image over real image

HOMOTOPY CONTINUATION

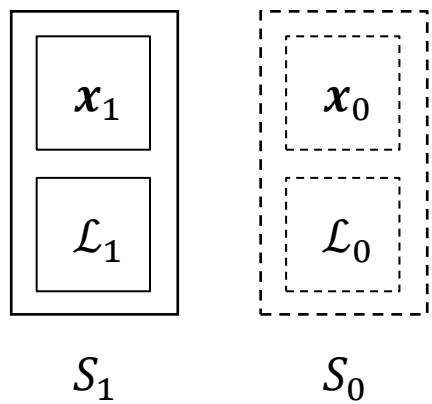
The real cameras $P_1, \dots, P_N$ to estimate are our problem unknowns $\mathbf{x}_1$, together with the measured lines set $\mathcal{L}_1$ form a starting system called S_1 .



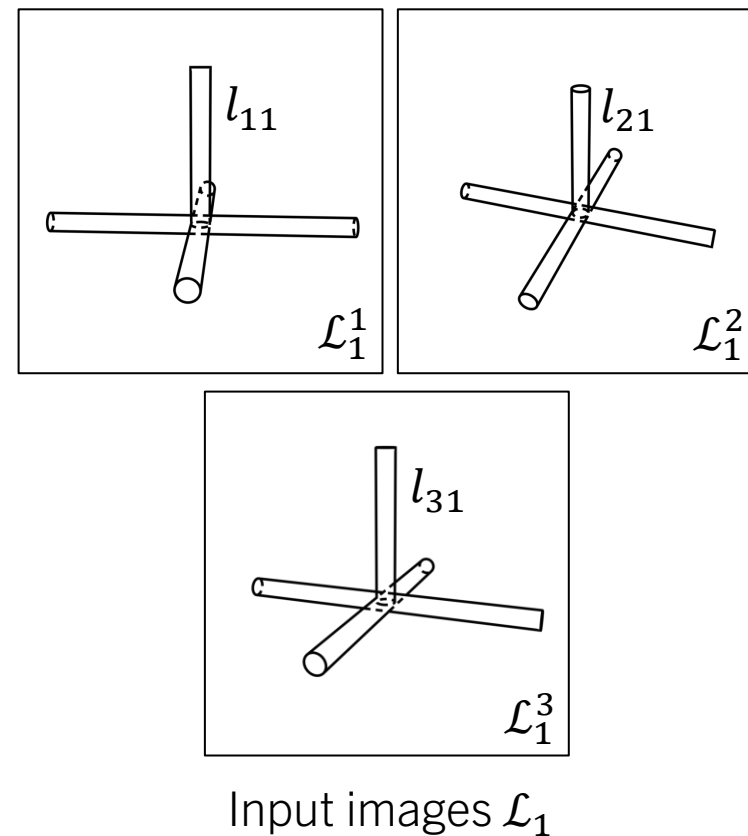
Sought cameras $\mathbf{x}_1$

HOMOTOPY CONTINUATION

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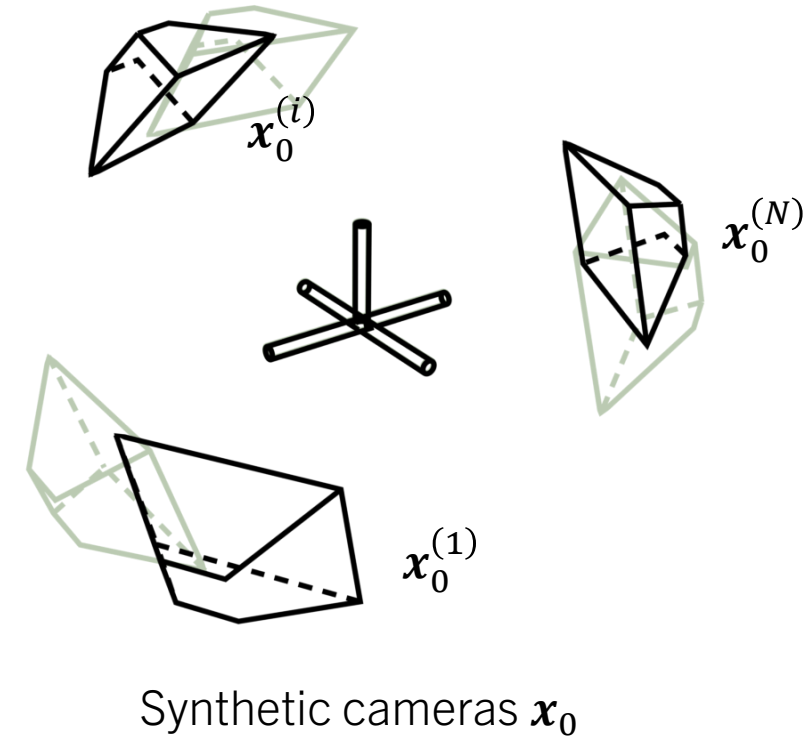
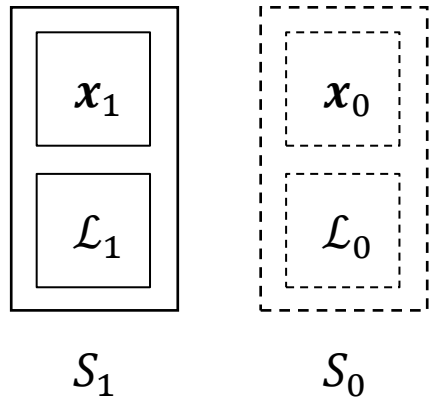
IMAGES



HOMOTOPY CONTINUATION

The real cameras $P_1, \dots, P_N$ to estimate are our problem unknowns $\mathbf{x}_1$, together with the measured lines set $\mathcal{L}_1$ form a starting system called S_1 .

We then generate a **random set of cameras** $P_1^R, \dots, P_N^R$ from which we project the very same 3D scene into a set of lines $\mathcal{L}_0$, **by construction** the camera set $\mathbf{x}_0$ is a **solution** of a new starting system S_0 .

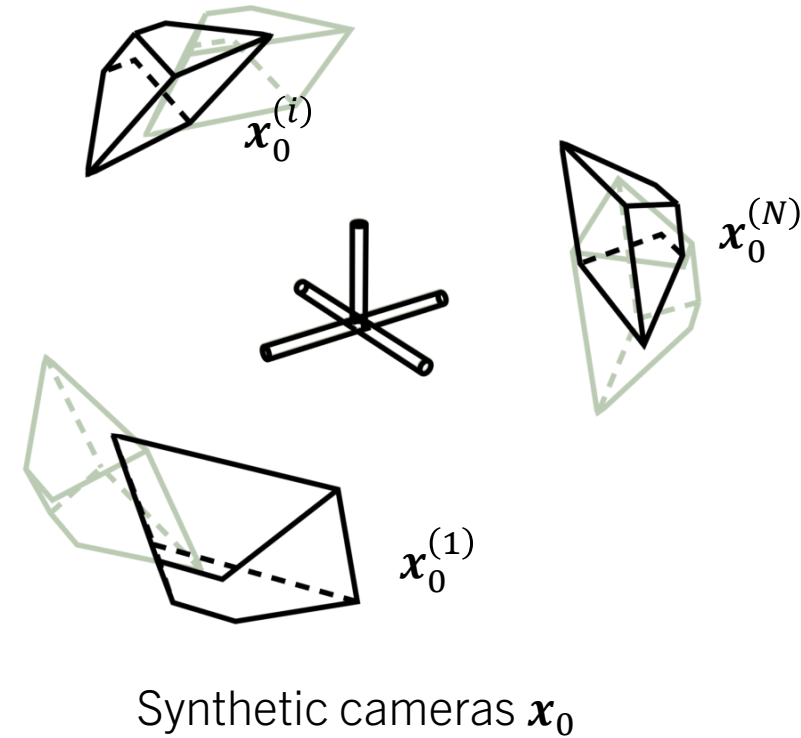
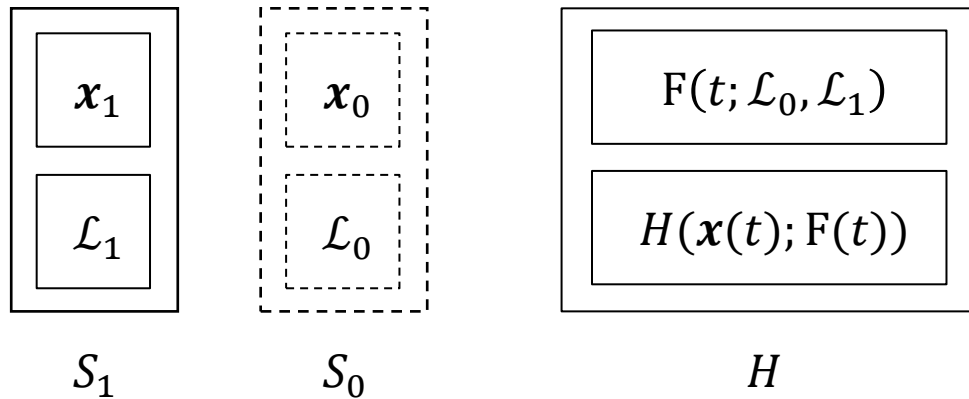


HOMOTOPY CONTINUATION

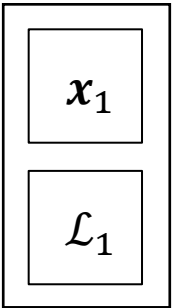
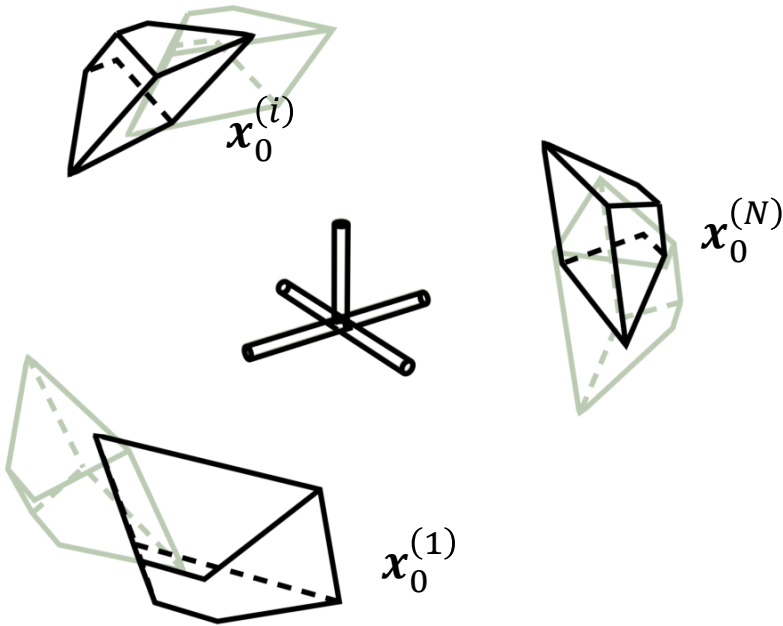
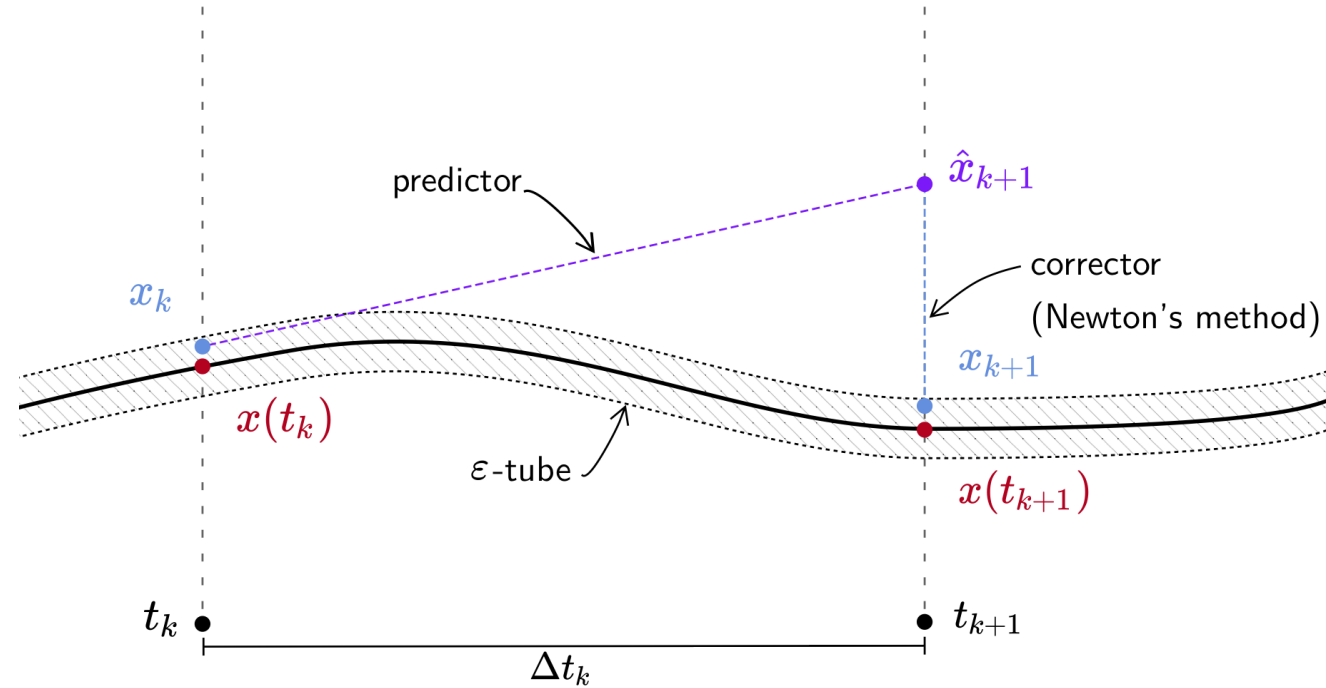
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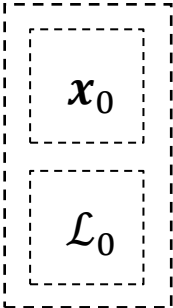
From this two systems S_0, S_1 a *homotopic function* $H = F(t; \mathcal{L}_0, \mathcal{L}_1)\mathcal{L}_0$ parametrized by $\mathcal{L}_0, \mathcal{L}_1$ is constructed.



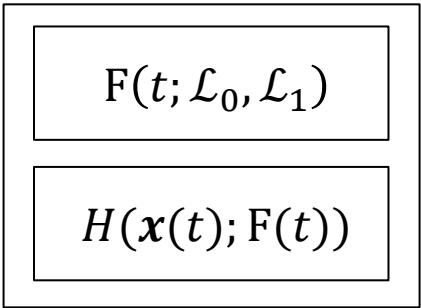
HOMOTOPY CONTINUATION



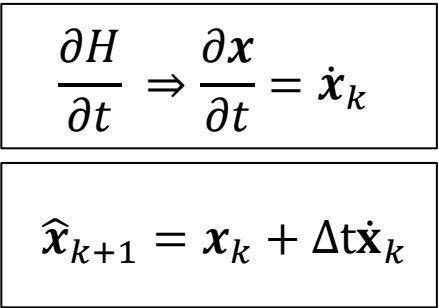
S_1



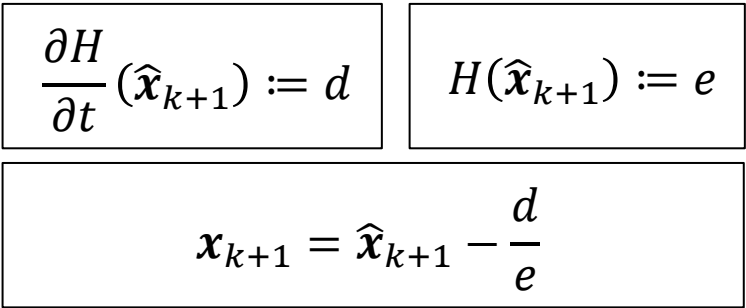
S_0



H

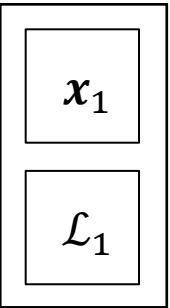
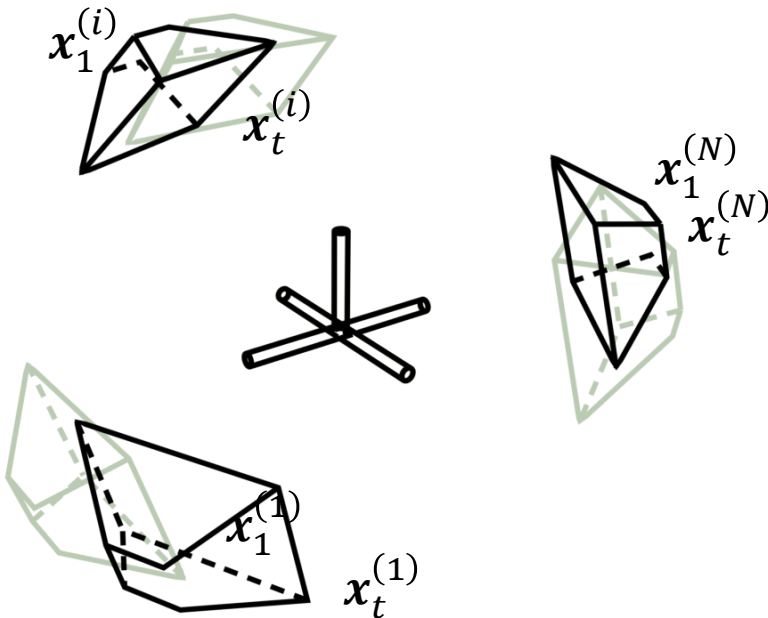
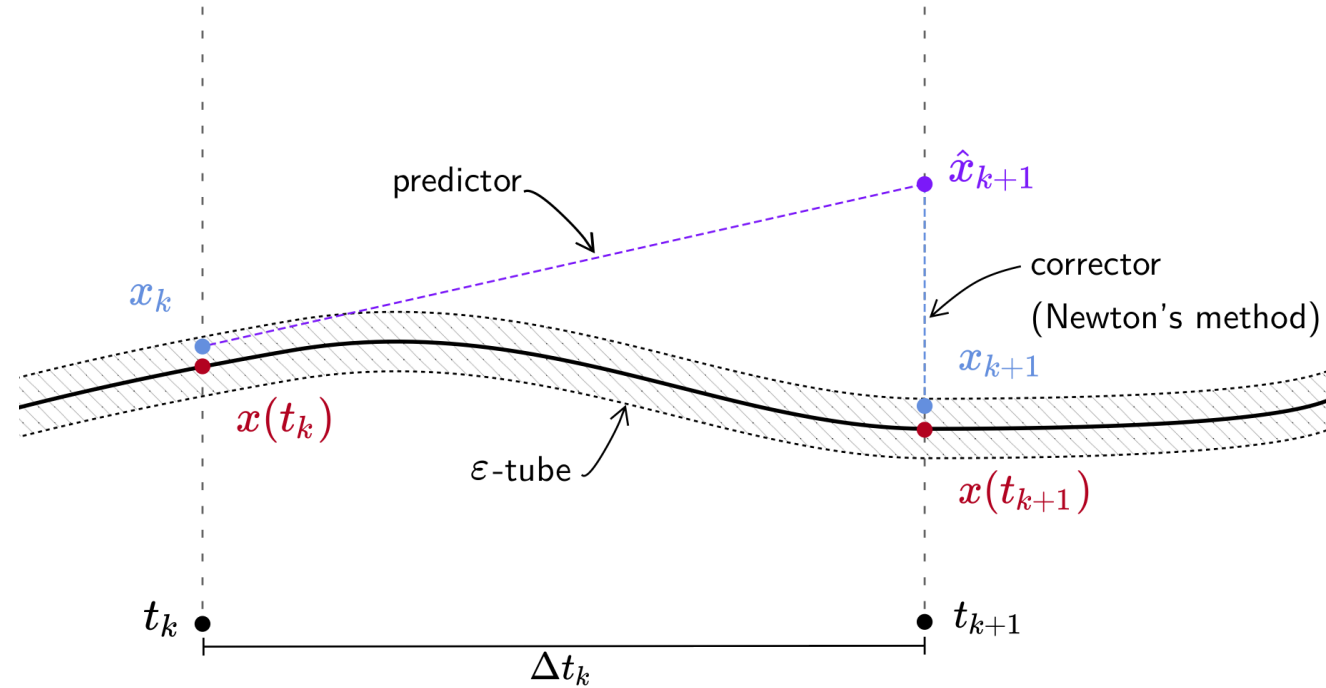


UPDATE

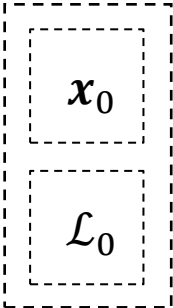


NEWTON CORRECTION

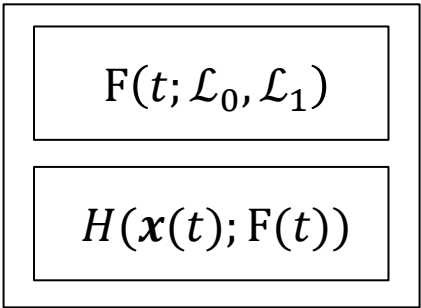
HOMOTOPY CONTINUATION



S_1



S_0



H

$$\frac{\partial H}{\partial t} \Rightarrow \frac{\partial x}{\partial t} = \dot{x}_k$$

$$\hat{x}_{k+1} = x_k + \Delta t \dot{x}_k$$

UPDATE

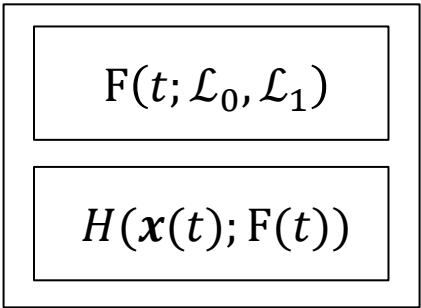
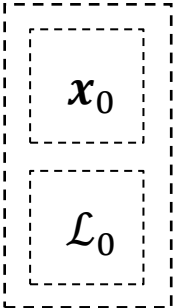
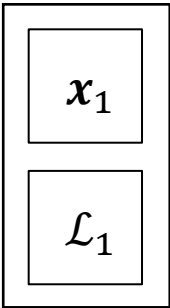
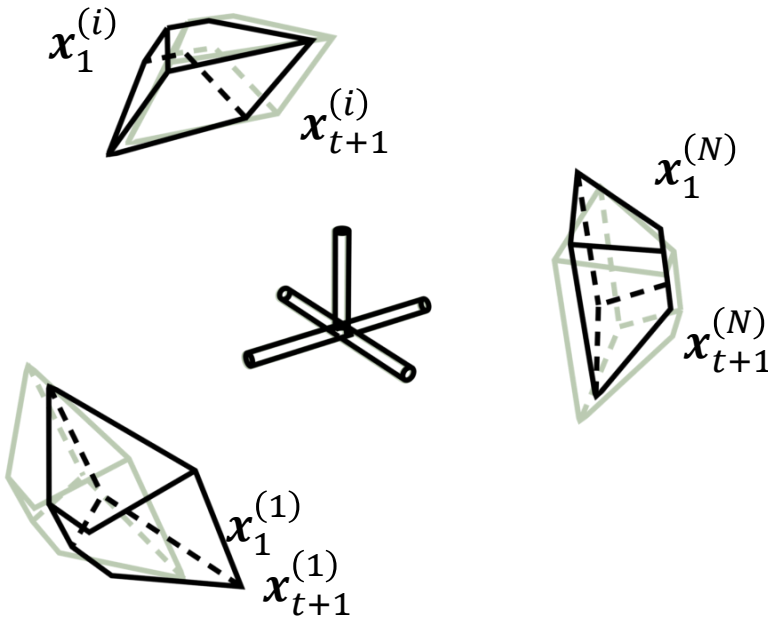
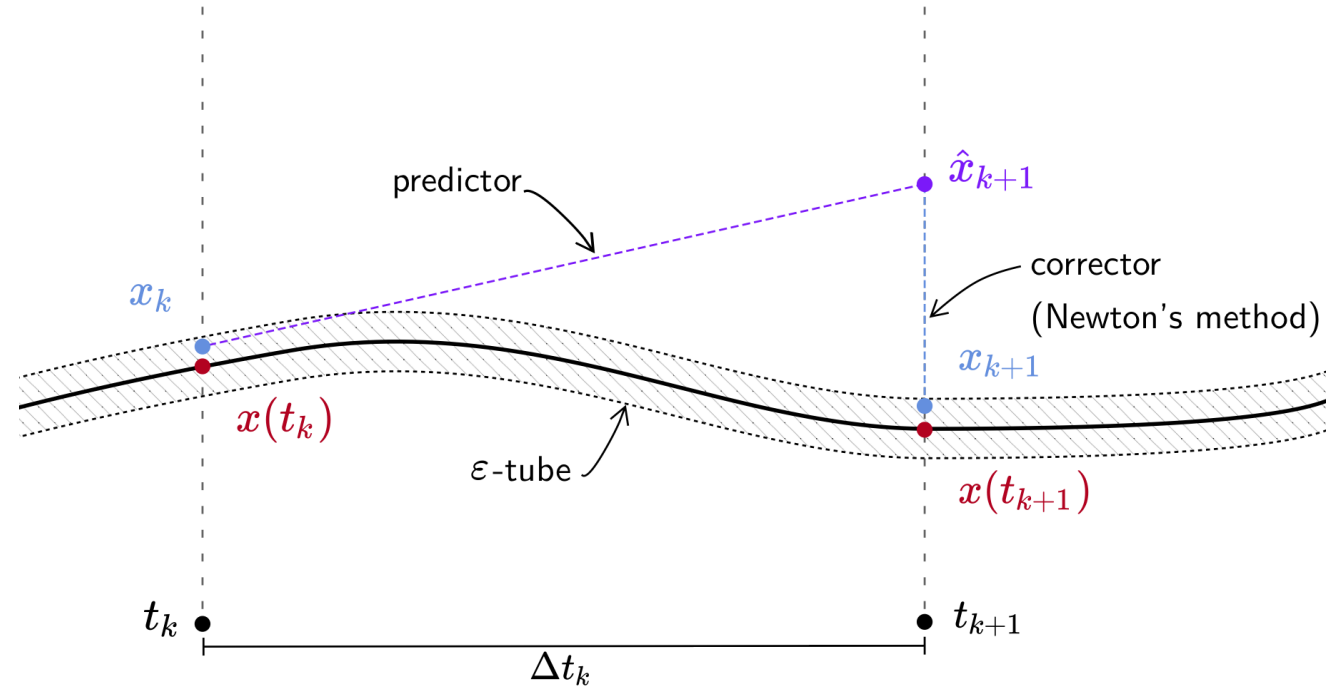
$$\frac{\partial H}{\partial t}(\hat{x}_{k+1}) := d$$

$$H(\hat{x}_{k+1}) := e$$

$$x_{k+1} = \hat{x}_{k+1} - \frac{d}{e}$$

NEWTON CORRECTION

HOMOTOPY CONTINUATION



$$\frac{\partial H}{\partial t} \Rightarrow \frac{\partial x}{\partial t} = \dot{x}_k$$

$$\hat{x}_{k+1} = x_k + \Delta t \dot{x}_k$$

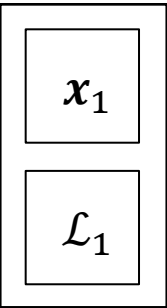
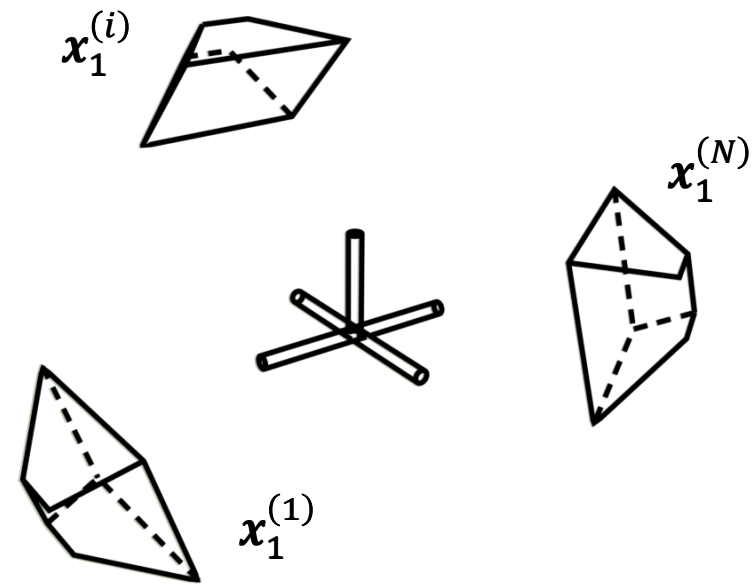
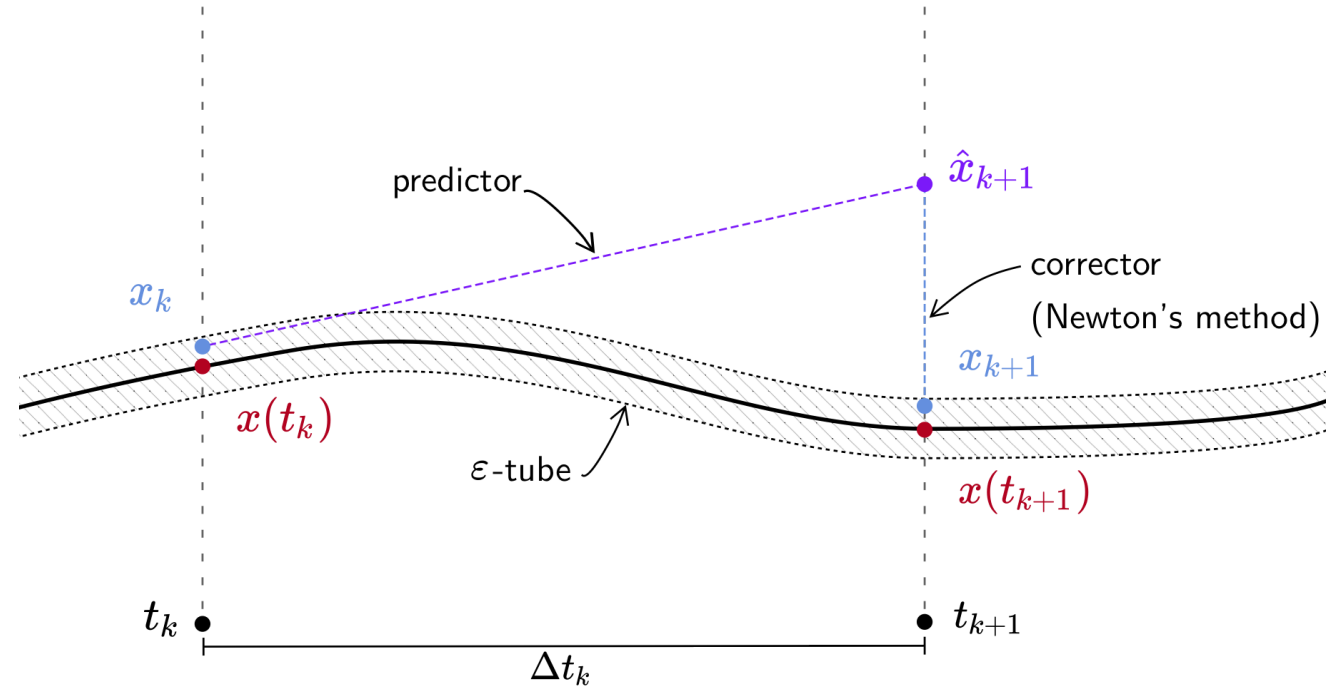
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$$x_{k+1} = \hat{x}_{k+1} - \frac{d}{e}$$

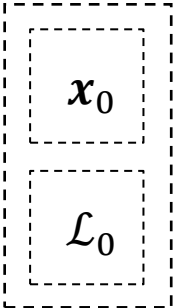
UPDATE

NEWTON CORRECTION

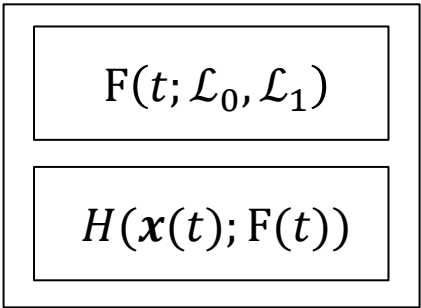
HOMOTOPY CONTINUATION



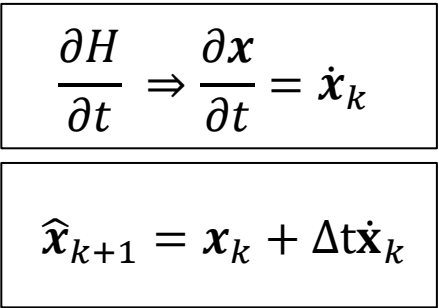
S_1



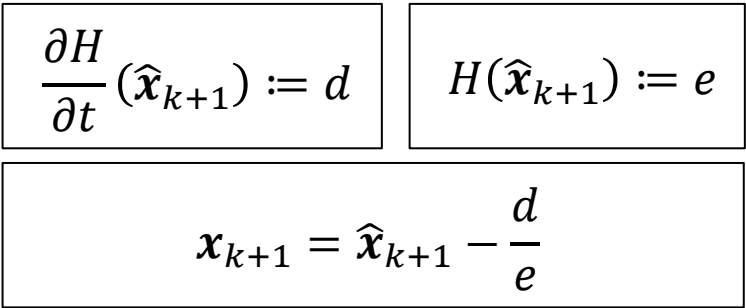
S_0



H



UPDATE



NEWTON CORRECTION

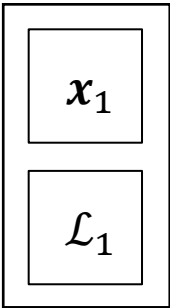
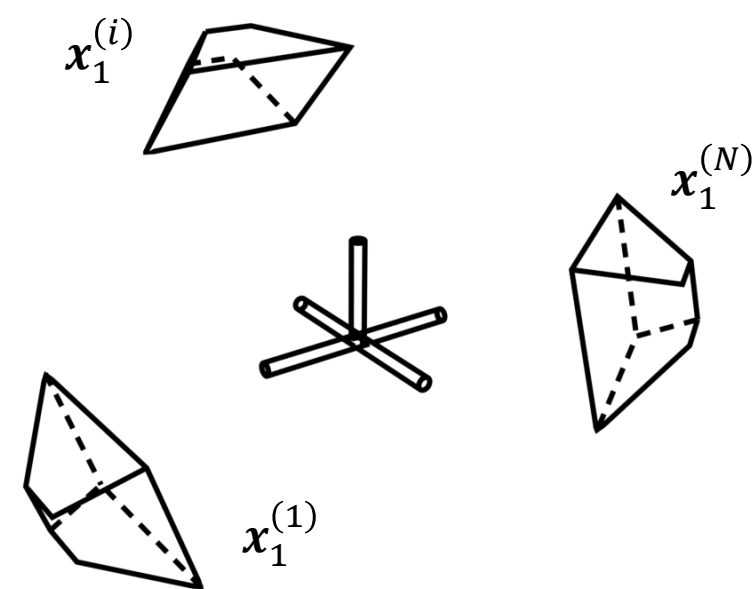
HOMOTOPY CONTINUATION

The real cameras $P_1, \dots, P_N$ to estimate are our problem unknowns $\mathbf{x}_1$, together with the measured lines set $\mathcal{L}_1$ form a starting system called S_1 .

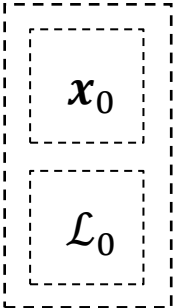
We then generate a **random set of cameras** $P_1^R, \dots, P_N^R$ from which we project the very same 3D scene into a set of lines $\mathcal{L}_0$, **by construction** the camera set $\mathbf{x}_0$ is a **solution** of a new starting system S_0 .

From this two systems S_0, S_1 a *homotopic function* $H = F(t; \mathcal{L}_0, \mathcal{L}_1)\mathcal{L}_0$ parametrized by $\mathcal{L}_0, \mathcal{L}_1$ is constructed.

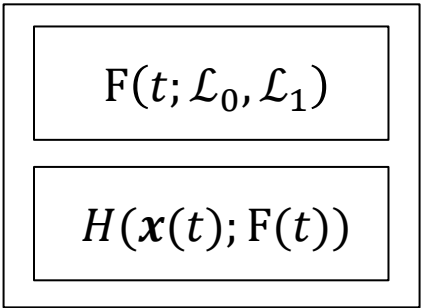
At the final iteration **we have found the unknown** $\mathbf{x}_1$.



S_1



S_0



H

$$\frac{\partial H}{\partial t} \Rightarrow \frac{\partial \mathbf{x}}{\partial t} = \dot{\mathbf{x}}_k$$

$$\hat{\mathbf{x}}_{k+1} = \mathbf{x}_k + \Delta t \dot{\mathbf{x}}_k$$

UPDATE

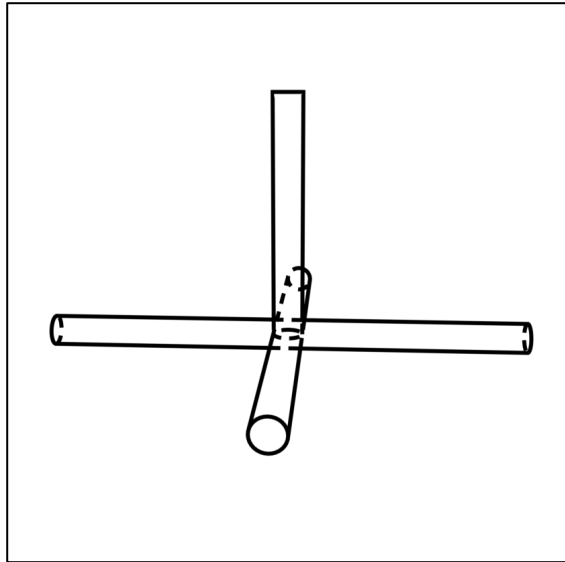
$$\frac{\partial H}{\partial t}(\hat{\mathbf{x}}_{k+1}) := d$$

$$H(\hat{\mathbf{x}}_{k+1}) := e$$

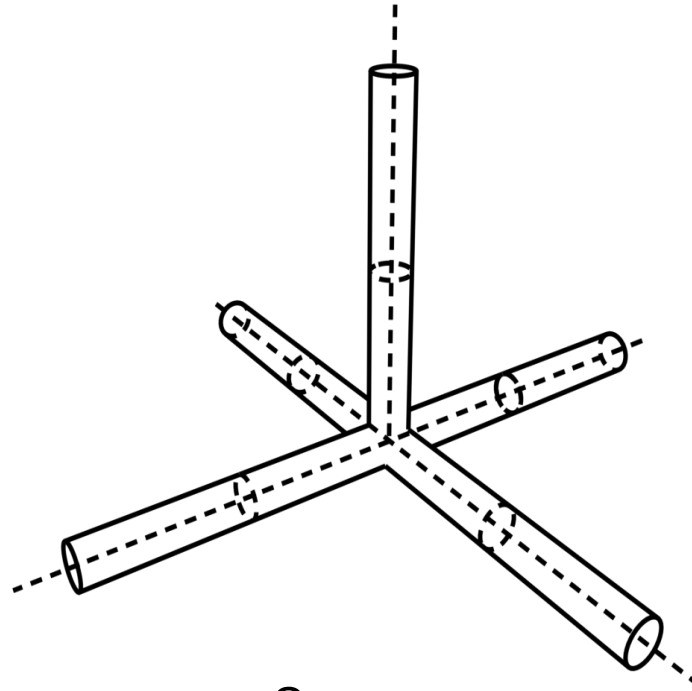
$$\mathbf{x}_{k+1} = \hat{\mathbf{x}}_{k+1} - \frac{d}{e}$$

NEWTON CORRECTION

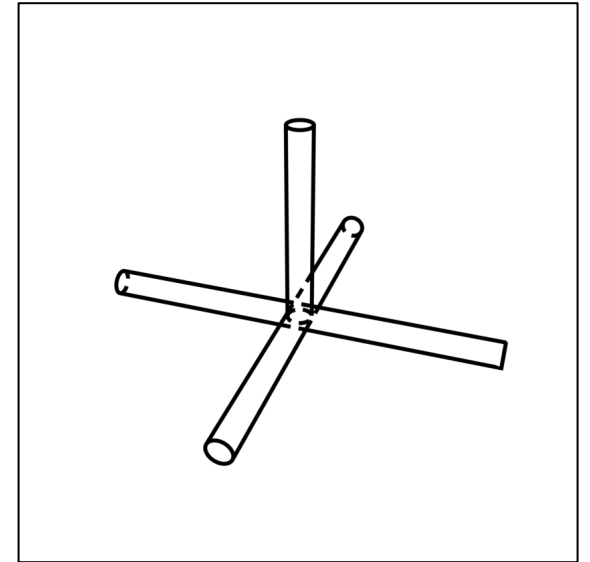
SILHOUETTE BASED PARAMETRIZATION



Synthetic image

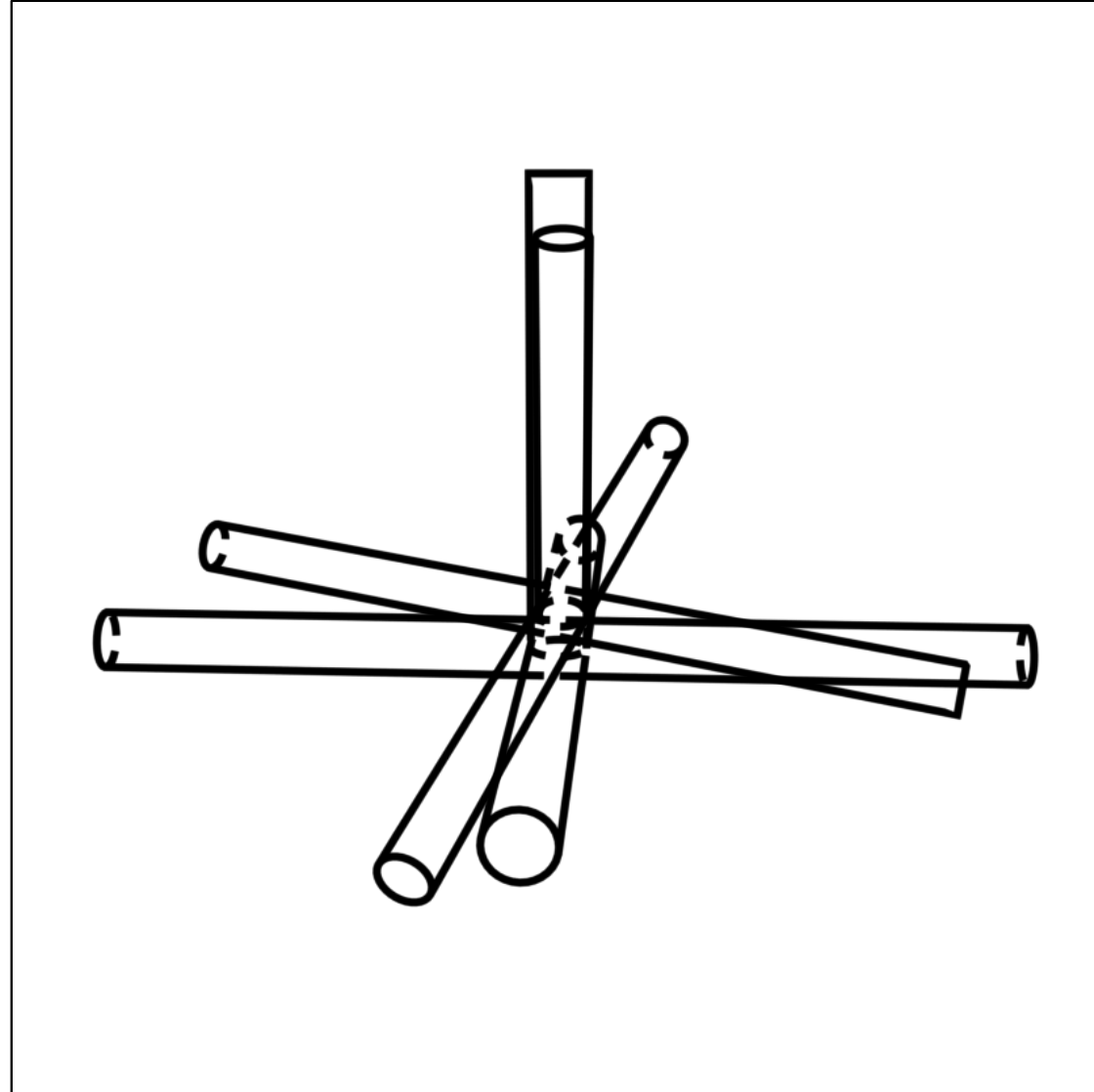


Scene



Real image

SILHOUETTE BASED PARAMETRIZATION

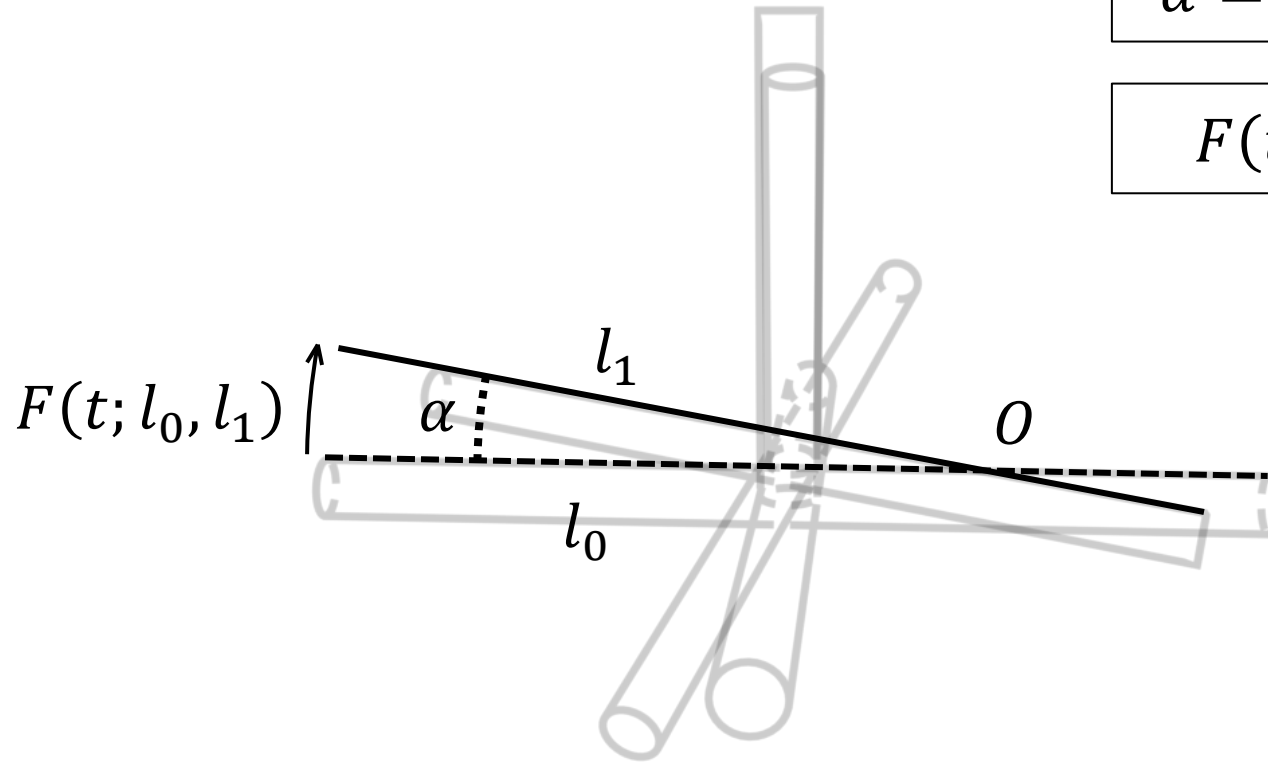


SILHOUETTE BASED PARAMETRIZATION

$$\alpha = \widehat{l_0 l_1}$$

$$O = l_0 \times l_1$$

$$F(t; l_0, l_1) = tT_{-O}R_{\alpha}T_O$$



ADDITIONAL CONTRIBUTIONS



SOLUTION GENERATION AND CORRECTION STEP

CORRECTION STEP BY FACTORIZATION

Homotopy continuation does not allow for inequality constraint, for this reason found camera parameters could be algebraically correct but geometrically infeasible.

To account for that we introduce a correction step achieved by factorizing KR . In equation (2) $l^h{}^T KR\mathbf{w} = 0$ we can see that if KR is a solution then also

$$l^T KCC^{-1}R\mathbf{w} = 0$$

is a solution for our system. C can't just be any matrix.

$C^{-1}R$ has to be a valid rotation, more formally has unitary determinant and be orthogonal

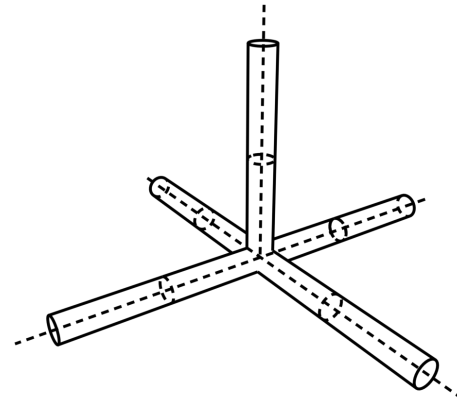
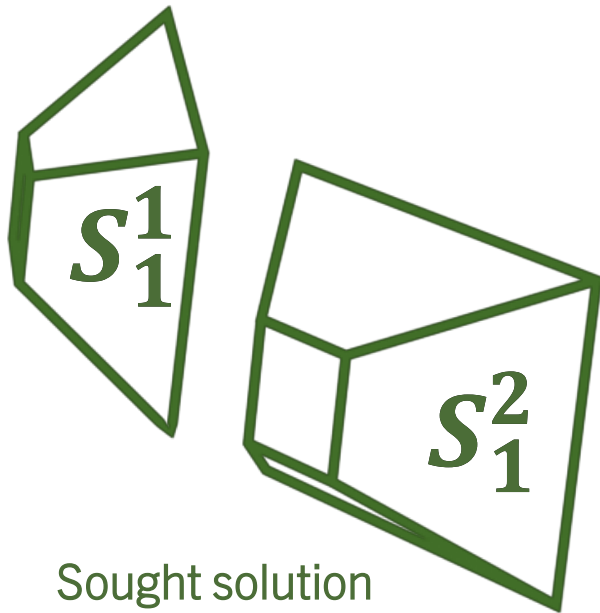
$$\begin{cases} \det(C^{-1}R) = \det(C^{-1}) = \pm 1 \\ (C^{-1}R)(C^{-1}R)^T = C^{-1}C^{-T} = I \end{cases}$$

$$C_{f_x} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_{f_y} = \begin{bmatrix} 1 & 2\alpha/f_x & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

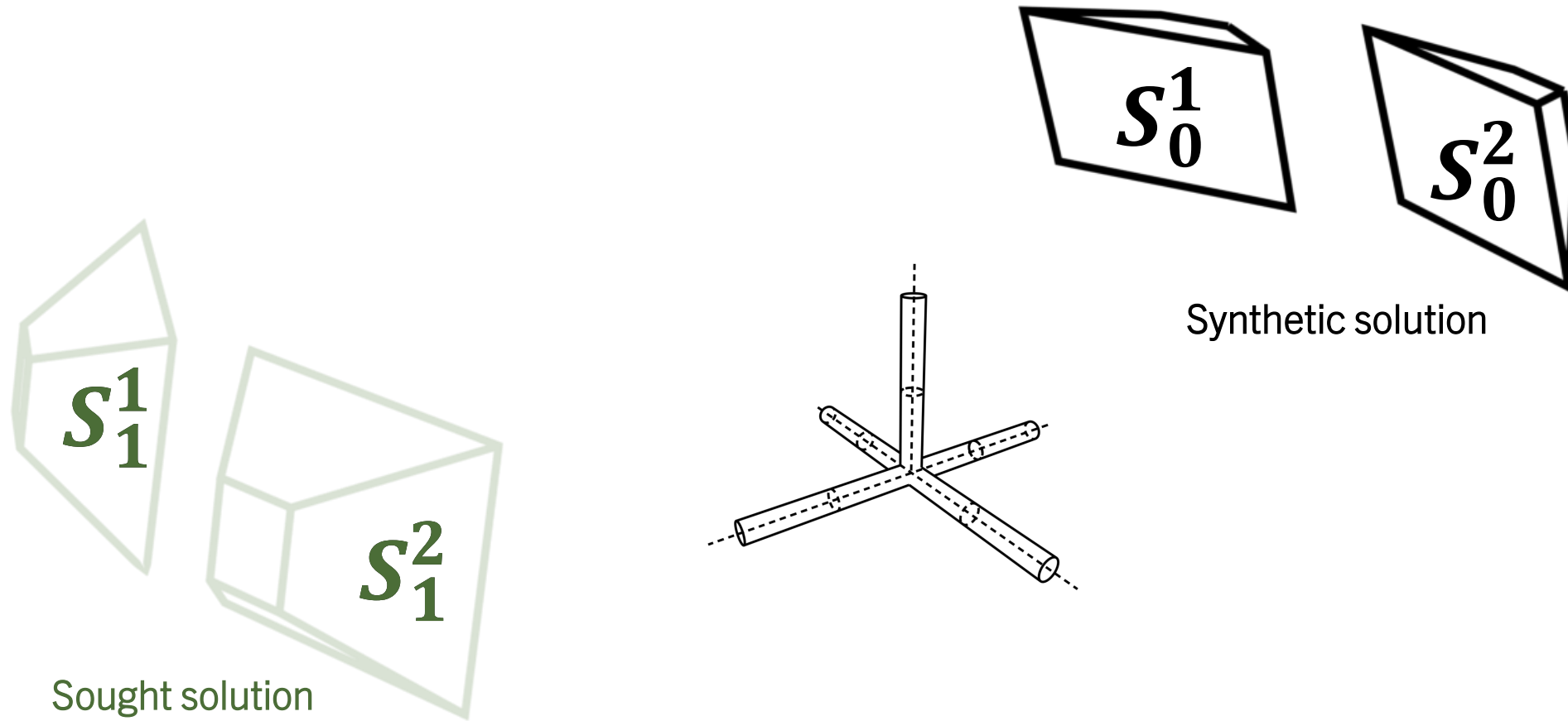
$$C_{sf_y} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION



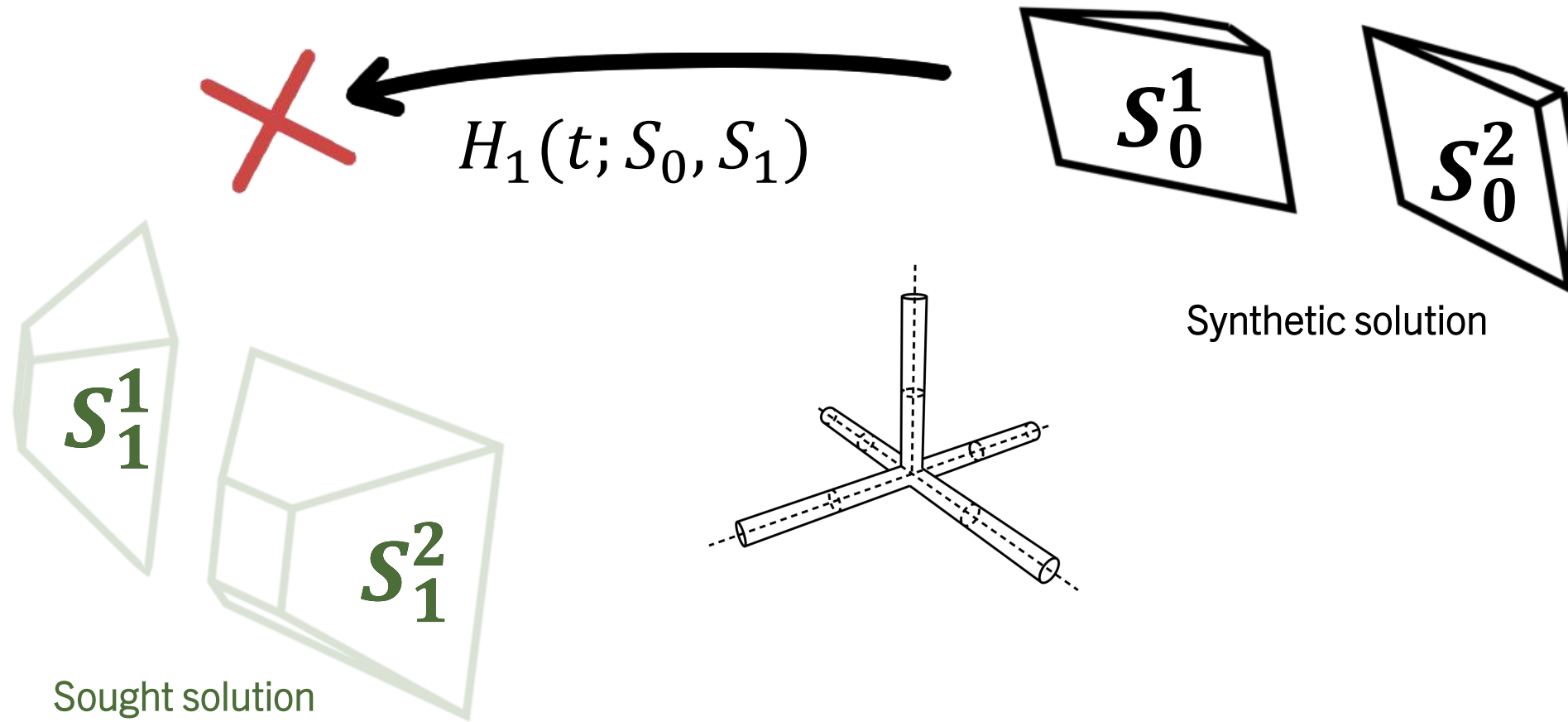
We start from two unknown cameras S_1^1, S_1^2 .

CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION



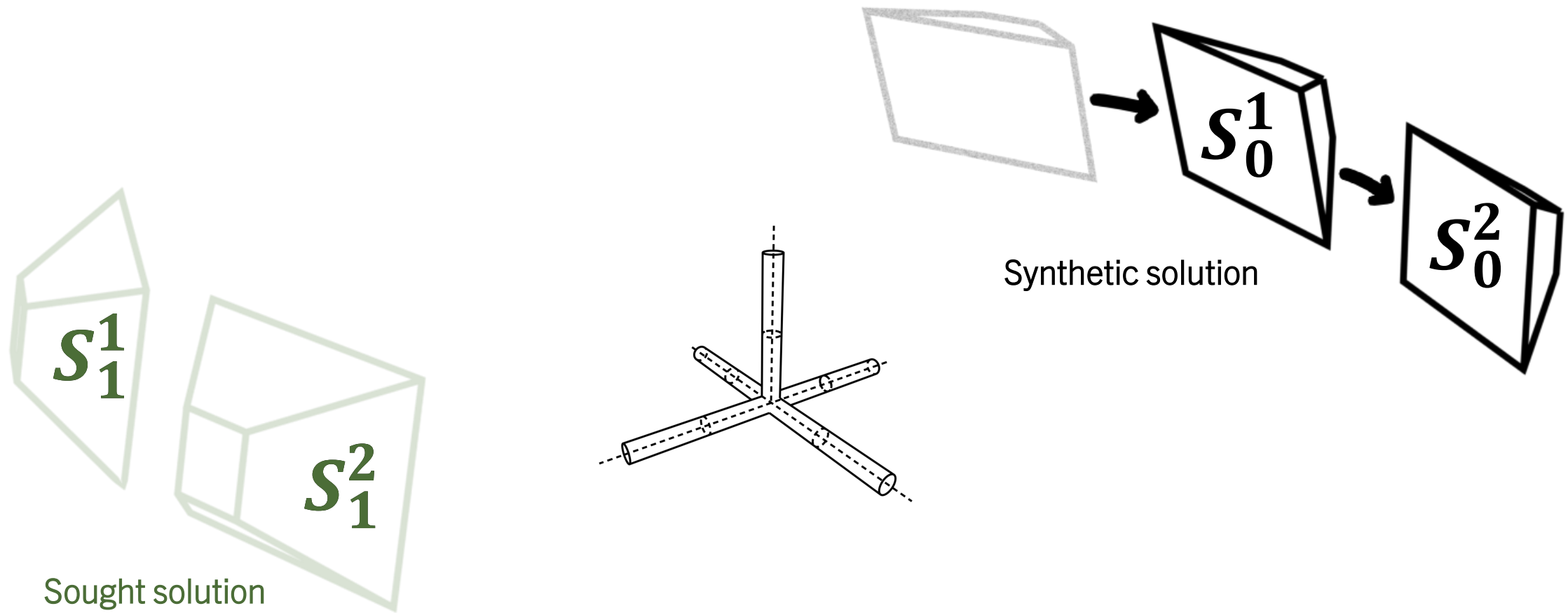
We generate a first synthetic camera S_0^1 in random position, and a random roto-translation offset $\mathbf{T}$. We apply $\mathbf{T}$ to get the second synthetic camera S_2^2 . This enforces local consistency between shots.

CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION



We try to run the homotopy H and find a path that leads to the initial solution, but it fails.

CAMERA RESECTIONING FROM CYLINDERS VIA HOMOTOPY CONTINUATION



The first camera S_0^1 of the pair is discarded, the second camera S_0^2 becomes the first one in the $k + 1$ iteration, and a new camera is added to the pair. If more than two cameras are present all are shifted by one, always discarding the first one and adding one.

EXPERIMENTS

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SYNTHETIC AND REAL

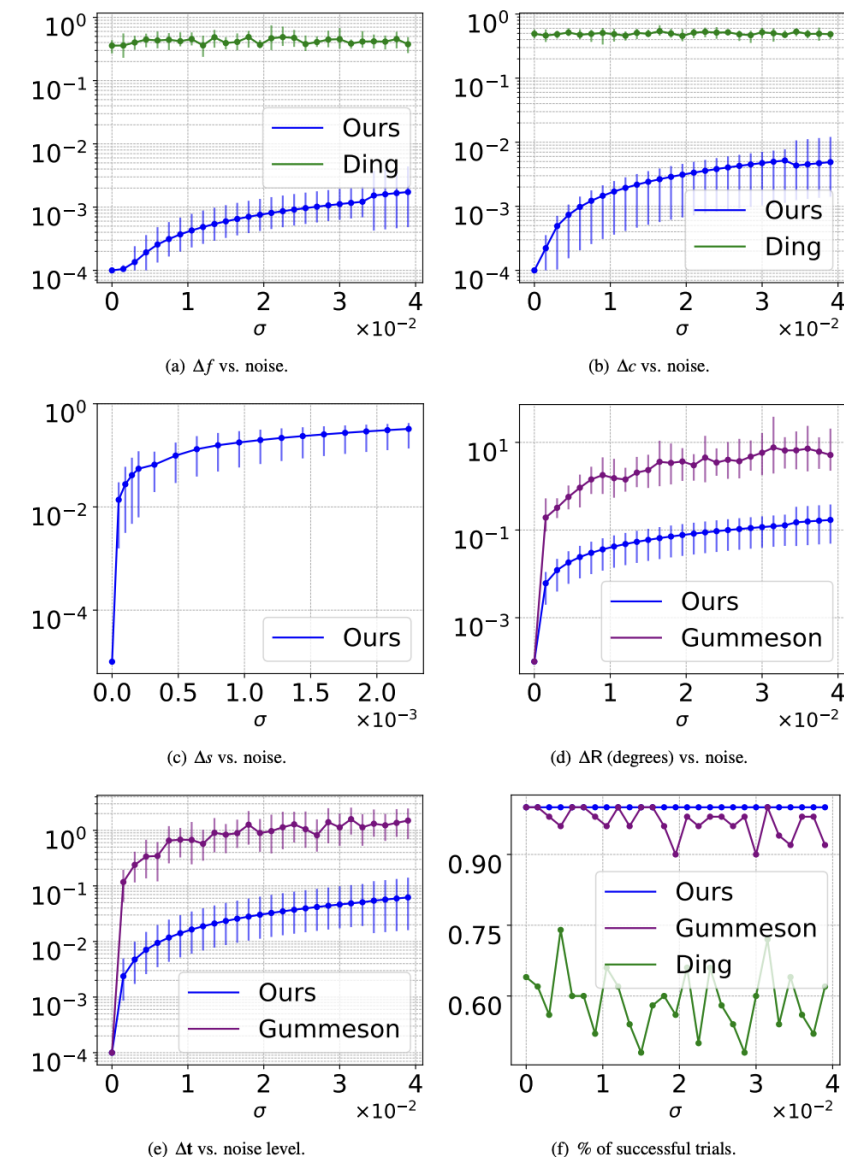
SYNTHETIC EXPERIMENTS

The experiment has been run for **increasing level of noise** from **0** to **$4e^{-2}$** . Each level of noise have been **tested** against **50 randomly drawn scenes**, and cameras.

All calibration parameters have been **assumed unknown**.

	Guo	Ding	Gummeson	Ours
Calibration	f ✗	K ✓	f ✗	K ✓
Pose estimation	R ✗	✗	R, t ✓	R, t ✓

✗: not addressed, ✗: partial output, ✓: fully solved



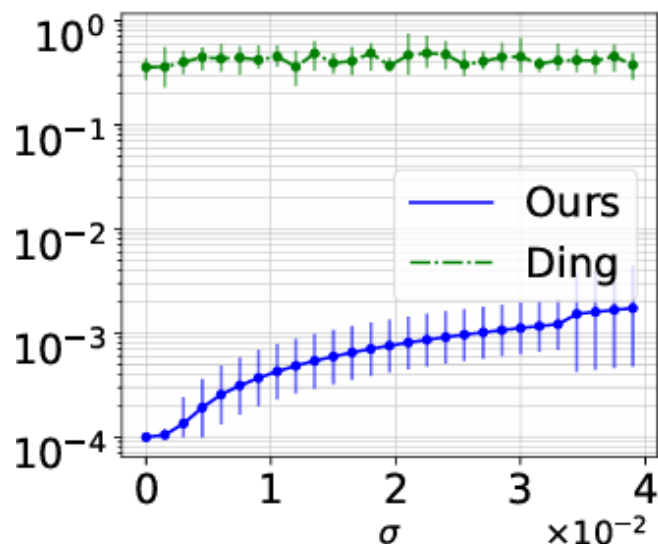
- (1) A. Gummeson, et al. Fast and efficient minimal solvers for quadric based camera pose estimation. In ICPR, 2022.
- (2) D.Ding, et al. Camera calibration based on free-placed single right cylindrical target. IEEE Sensors Journal, 2024.

CALIBRATION EXPERIMENTS

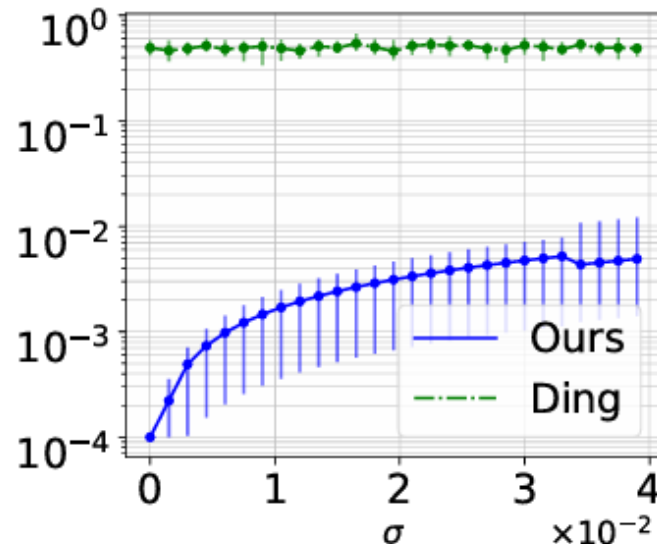
On camera calibration the proposed method has been tested against *Ding et al.* [1]. Our method performs order of magnitudes better than the competition. As noted on the right our method is the only one capable of estimating the skew parameter γ .

	Guo	Ding	Gummesson	Ours
Calibration	f ✗	K ✓	f ✗	K ✓
Pose estimation	R ✗	✗	R, t ✓	R, t ✓

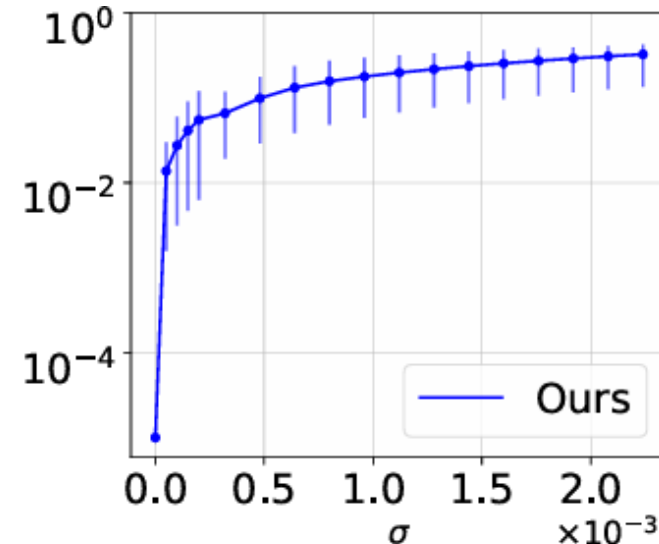
✗: not addressed, ✗: partial output, ✓: fully solved



(a) Δf vs. noise.



(b) Δc vs. noise.



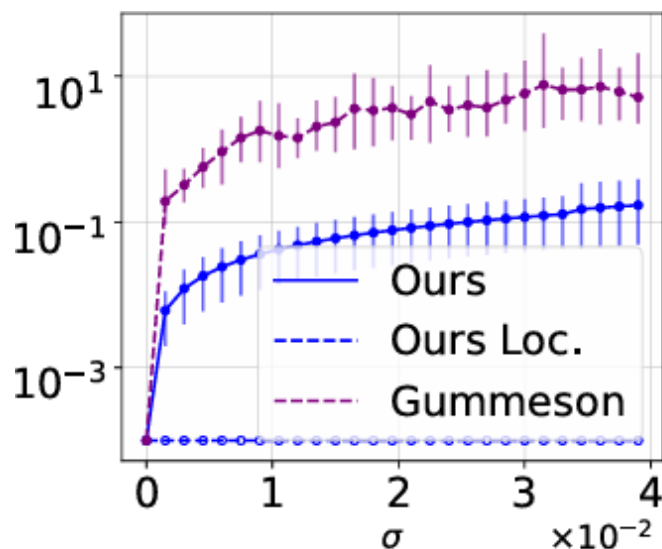
(c) Δs vs. noise.

CALIBRATION EXPERIMENTS

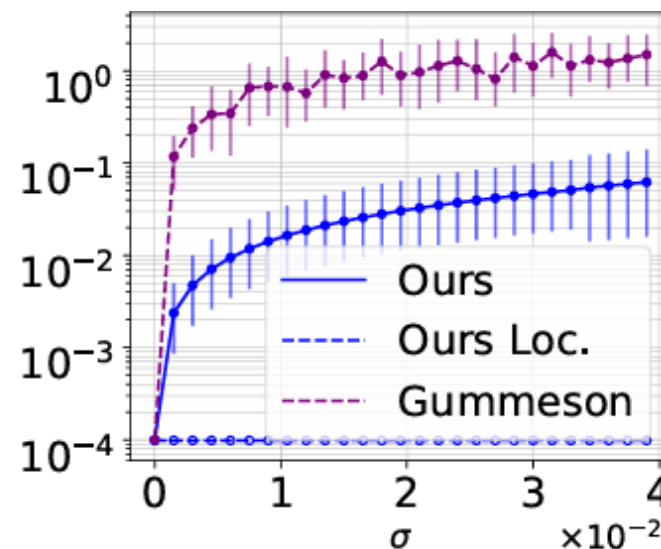
On camera localization the proposed method has been tested against *Gummeson et al.* [1]. It has been tested while also performing calibration (middle blue line), and also tested in the calibrated case in which performs perfectly (bottom line).

	Guo	Ding	Gummeson	Ours
Calibration	f ✗	K ✓	f ✗	K ✓
Pose estimation	R ✗	✗	R, t ✓	R, t ✓

✗: not addressed, ✗: partial output, ✓: fully solved



(d) ΔR (degrees) vs. noise.



(e) Δt vs. noise level.

REAL EXPERIMENTS

In terms of reprojection error (right figure), Gummeson (top), and our method (bottom) performs comparably to one another. Notably, our does so while also estimating camera intrinsic parameters.

Scene	f_x	f_y	c_x	c_y
Pipes	2563.36	2924.82	518.66	1242.12
Pipes (gt)	1500.00	2666.67	540.00	960.00
Pipes (Δ)	42.48%	8.83%	4.12%	22.71%
Roller coaster	1741.12	1711.91	918.0	538.0
Roller coaster (gt)	1749.83	1751.29	960.0	540.0
Roller coaster (Δ)	0.50%	2.30%	4.58%	0.37%
Markers	1334.72	1234.91	848.93	960.00
Markers (gt)	1632.60	1605.51	540.00	960.00
Markers (Δ)	22.32%	30.01%	36.39%	0.00%
Mean Δ	21.77%	13.71%	15.03%	7.69%

Absolute values and relative error on estimated intrinsic parameters for real scens. Roller coaster scene on the right highlighted in blue.



GUMMESON

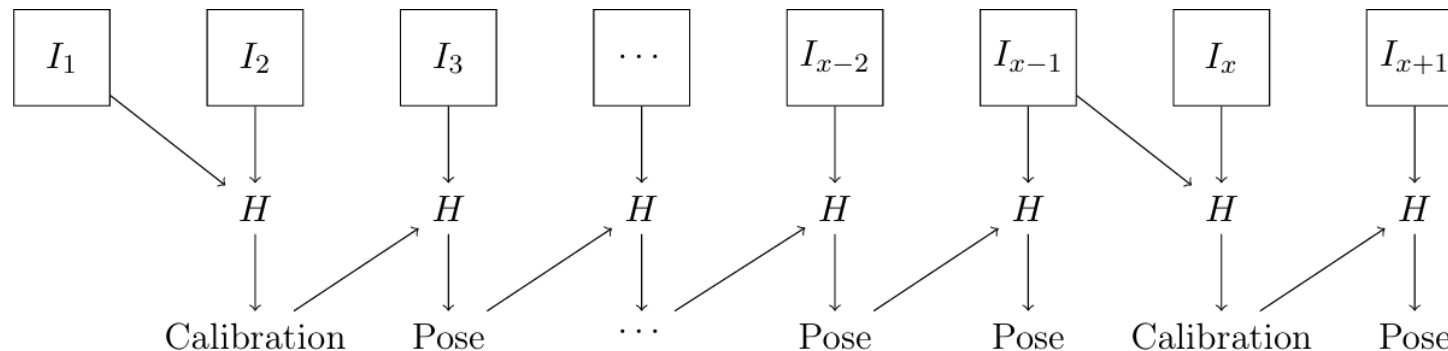
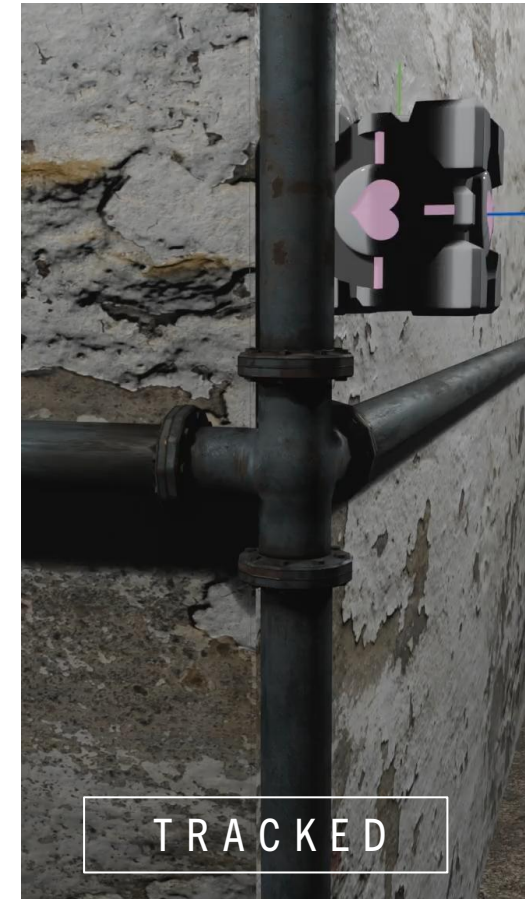
OURS

POSSIBLE APPLICATION: SPECIAL EFFECTS

The **flexibility** of our method allows to **seamlessly transition** between a **calibration** and a **localization** task.

This allows for a localization/calibration **feedback loop** in which **calibration** is performed only **every N-th frame** or if the **localization procedure fails**.

This allows for fast tracking while the camera is moving while also preserving local consistency



H is our homotopy function. It can be used with **2 frames** to perform a **calibration+localization** step. Iterating the procedure is possible to provide the **previous pose**, and a **new frame** to estimate the next pose.

CONCLUSIONS

Our method is the first to **jointly estimate camera intrinsic K parameters and camera poses $[R_i | \mathbf{t}_i]$** .

Our **method flexibility** opens up interesting scenarios where mixed steps of calibration and pose are needed.

Performance exceed state of the art method for cylinders.

A **novel correction step** applicable to all problems in which **K** and **R** are jointly estimated.

FUTURE WORK

Research practical applications of our method. **Seamlessly transitioning between calibration and pose estimation** opens up new possibilities.

Restricting the scene to **reduce the amount of input required**.